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Reduction of common mode emission of an electrical power converter

Abstract

A power system including a power converter system and an electric machine is provided. In one aspect, the power converter system has first and second switching elements. The electric machine includes a first multiphase winding electrically coupled with the first switching elements and a second multiphase winding electrically coupled with the second switching elements. The first and second multiphase windings are arranged and configured to operate electrically opposite in phase with respect to one another. One or more processors control the first switching elements to generate first pulse width modulated (PWM) signals based on received voltage commands to render a first common mode signal and also control the second switching elements to generate second PWM signals based on received voltage commands to render a second common mode signal. The rendered first and second common mode signals have the same or similar waveform with opposite polarity with respect to one another.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/507,866, filed Oct. 22, 2021, titled “Reduction of Common Mode Emission of an Electrical Power Converter,” which is incorporated herein by reference in its entirety for all purposes.

FIELD

(1) The present subject matter relates generally to electrical power systems, such as electrical power systems for aircraft.

BACKGROUND

(2) A conventional commercial aircraft generally includes a fuselage, a pair of wings, and a propulsion system that provides thrust. The propulsion system typically includes at least two aircraft engines, such as turbofan jet engines. Each turbofan jet engine is typically mounted to a respective one of the wings of the aircraft, such as in a suspended position beneath the wing separated from the wing and fuselage.

(3) Hybrid-electric propulsion systems are being developed to improve an efficiency of conventional commercial aircraft. Some hybrid electric propulsion systems include one or more electric machines each being mechanically coupled with a rotating component of one of the aircraft engines. The electric machines can each have an associated power converter electrically connected thereto. The inventors of the present disclosure have developed various systems and methods to improve hybrid electric propulsion systems, and more generally, power systems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

(2) FIG. **1** provides a schematic top view of an aircraft having a hybrid-electric propulsion system according to various exemplary embodiments of the present disclosure;

(3) FIG. **2** provides a schematic cross-sectional view of one of the hybrid-electric propulsors of the aircraft of FIG. **1**;

(4) FIG. **3** provides a schematic cross-sectional view of a hybrid-electric propulsor that may be implemented with the aircraft of FIG. **1**;

(5) FIG. **4** provides a schematic cross-sectional view of an electric machine embedded in a gas turbine engine of the hybrid-electric propulsor of FIG. **2**;

(6) FIG. **5** provides a schematic cross-sectional view of an electric machine having an inner-rotor configuration and being embedded in a gas turbine engine according to various exemplary embodiments of the present disclosure;

(7) FIG. **6** provides a schematic view of an electrical power system according to one example embodiment of the present disclosure;

(8) FIG. **7** provides a schematic view of the multiphase windings of the electric machine of FIG. **6**;

(9) FIG. **8** provides a schematic view of first switching elements of a power converter system electrically coupled with a first multiphase winding of the power system of FIG. **6**;

(10) FIG. **9** provides a topology of a control system of the power system of FIG. **6**;

(11) FIGS. **10**A through **10**D provide various graphs depicting voltage signals as a function of time;

(12) FIGS. **11**A and **11**B provide graphs depicting an example manner in which a pulse width modulated signal associated with a first multiphase winding of the electric machine of FIG. **6** can be generated;

(13) FIGS. **12**A and **12**B provide graphs depicting an example manner in which a pulse width modulated signal associated with a second multiphase winding of the electric machine of FIG. **6** can be generated;

(14) FIG. **13** depicts a simplified graph depicting various signals as a function of time;

(15) FIG. **14** provides a schematic view of an electrical power system according to one example embodiment of the present disclosure;

(16) FIG. **15** provides a topology of a control system of the power system of FIG. **14**;
(17) FIG. **16** provides a schematic view of an electrical power system according to one example embodiment of the present disclosure;
(18) FIG. **17** provides a schematic view of an electrical power system according to one example embodiment of the present disclosure;
(19) FIG. **18** provides a schematic view of an electrical power system according to one example embodiment of the present disclosure;
(20) FIG. **19** provides a flow diagram for a method of operating an electrical power system according to one example embodiment;
(21) FIG. **20** provides a schematic view of second switching elements of the power converter system electrically coupled with a second multiphase winding of the power system of FIG. **6**, wherein some of the second switching elements have experienced a failure condition;
(22) FIGS. **21**A through **21**C provide various graphs depicting voltage signals as a function of time;
(23) FIG. **22** provides a schematic view of second switching elements of the power converter system electrically coupled with a second multiphase winding of the power system of FIG. **6**, wherein one of the second switching elements has experienced a failure condition;
(24) FIGS. **23**A through **23**D provide various graphs depicting voltage signals as a function of time;
(25) FIGS. **24**A through **24**E provide various graphs depicting voltage signals as a function of time;
(26) FIGS. **25**A through **25**E provide various graphs depicting voltage signals as a function of time;
(27) FIGS. **26**A, **26**B, and **26**C each show a graph depicting voltage signals as a function of time;
(28) FIG. **27** provides a flow diagram for a method of operating an electrical power system according to one example embodiment;
(29) FIG. **28** provides a schematic cross-sectional view of one example axial flux rotating electric machine according to an example embodiment of the present disclosure;
(30) FIG. **29** provides a schematic cross-sectional view of another example axial flux rotating electric machine according to an example embodiment of the present disclosure; and
(31) FIG. **30** provides an example computing system according to example embodiments of the present disclosure.

DETAILED DESCRIPTION

(32) Reference will now be made in detail to present embodiments of the invention, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the invention.
(33) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.
(34) As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.
(35) The terms “forward” and “aft” refer to relative positions within a gas turbine engine or vehicle, and refer to the normal operational attitude of the gas turbine engine or vehicle. For example, with regard to a gas turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.
(36) The terms “upstream” and “downstream” refer to the relative direction with respect to a flow in a pathway. For example, with respect to a fluid flow, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows. However, the terms “upstream” and “downstream” as used herein may also refer to a flow of electricity.

(37) The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.
(38) Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 1, 2, 4, 5, 10, 15, or 20 percent margin in either individual values, range(s) of values and/or endpoints defining range(s) of values.
(39) Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.
(40) Electrical power systems, such as those found in aircraft hybrid-electric propulsion systems, can employ an electric machine and a power converter system electrically connected thereto. Due to ever increasing requirements for aviation electrical power systems to increase power distribution voltage, increase power level and consequent emission paths, and use of efficient high-speed power semiconductors, there is an increased need for mitigation of common mode emissions. Common mode emissions may introduce voltage stresses and parasitic current via electromechanical interfaces.
(41) Accordingly, the inventors of the present disclosure have developed architectures and control schemes that may reduce common mode emissions and associated electromagnetic interference in electrical power systems having an electric machine and a power converter system electrically coupled thereto. In one example aspect, polyphase or multiphase windings of an electric machine are arranged to operate under complementary excitation and PWM excitations are synthesized at the power converter system to reduce common mode emissions by cancelation.
(42) Particularly, in one example embodiment, a power system including a power converter system and an electric machine is provided. The power converter system has first switching elements and second switching elements. The electric machine includes a first multiphase winding electrically coupled with the first switching elements and a second multiphase winding electrically coupled with the second switching elements. The first and second multiphase windings are arranged and are configured to operate electrically opposite in phase with respect to one another. That is, the first multiphase winding and the second multiphase winding are electrically out-of-phase with respect to one another by one hundred eighty degrees (180°).
(43) One or more processors of the power system can receive voltage commands associated with the first multiphase winding and voltage commands associated with the second multiphase winding. The one or more processors can control the first switching elements to generate first pulse width modulated (PWM) signals based at least in part on the voltage commands associated with the first multiphase winding. The generated first PWM signals effectively render a first common mode signal. Likewise, the one or more processors can control the second switching elements to generate second PWM signals based at least in part on the voltage commands associated with the second multiphase winding. The generated second PWM signals effectively render a second common mode signal.
(44) Notably, in some instances, the second common mode signal has a same or similar waveform with opposite polarity with respect to the first common mode signal. The first and second common mode signals have the same or similar waveform because the first and second multiphase windings

are electrically opposite in phase with respect to one another. The polarity of the common mode signals are made opposite one another due to the one or more processors changing the polarity of the first or second PWM signals in some regard. For instance, the polarity of the second PWM signals can be changed by shifting a carrier signal to which the voltage commands associated with the second multiphase winding are compared by one hundred eighty degrees (180°) with respect to a carrier signal to which the voltage commands associated with the first multiphase winding are compared.

(45) As the first and second common mode signals have the same or similar waveform and opposite polarity, common mode emissions can be canceled or reduced. Advantages and benefits may be realized by cancelation or reduction of common mode emissions. For instance, the need for EMI filters can be eliminated or at least one or more EMI filters can be reduced in size. This may be advantageous for weight sensitive applications, such as aviation applications. Cancelation or reduction of common mode emissions can also reduce shaft voltage and bearing currents, thereby potentially: reducing bearing stress, eliminating the need for a shaft grounding brush, eliminating the need of a bearing insulation sleeve or ceramic bearing, and/or reducing leakage current through shaft loads, such as gears or sensors.

(46) Moreover, the architectures and control schemes disclosed herein may also be utilized advantageously to cancel common mode emissions even for power systems that do not include a wire dedicated to facilitating cancelation of common mode emissions. Some conventional systems include such dedicated wires. In this regard, the inventive aspects disclosed herein may provide common mode emission cancelation for a wide variety of systems, including for example, isolated neutral systems. Other benefits and advantages may be realized as well.

(47) FIG. **1** provides a schematic top view of an exemplary aircraft **100** as may incorporate one or more inventive aspects of the present disclosure. As shown in FIG. **1**, for reference, the aircraft **100** defines a longitudinal direction L1 and a lateral direction L2. The lateral direction L2 is perpendicular to the longitudinal direction L1. The aircraft **100** also defines a longitudinal centerline **114** that extends therethrough along the longitudinal direction L1. The aircraft **100** extends between a forward end **116** and an aft end **118**, e.g., along the longitudinal direction L1.

(48) As depicted, the aircraft **100** includes a fuselage **112** that extends longitudinally from the forward end **116** of the aircraft **100** to the aft end **118** of the aircraft **100**. The aircraft **100** also includes an empennage **119** at the aft end **118** of the aircraft **100**. In addition, the aircraft **100** includes a wing assembly including a first, port side wing **120** and a second, starboard side wing **122**. The first and second wings **120**, **122** each extend laterally outward with respect to the longitudinal centerline **114**. The first wing **120** and a portion of the fuselage **112** together define a first side **124** of the aircraft **100** and the second wing **122** and another portion of the fuselage **112** together define a second side **126** of the aircraft **100**. For the embodiment depicted, the first side **124** of the aircraft **100** is configured as the port side of the aircraft **100** and the second side **126** of the aircraft **100** is configured as the starboard side of the aircraft **100**.

(49) The aircraft **100** includes various control surfaces. For this embodiment, each wing **120**, **122** includes one or more leading edge flaps **128** and one or more trailing edge flaps **130**. The aircraft **100** further includes, or more specifically, the empennage **119** of the aircraft **100** includes a vertical stabilizer **132** having a rudder flap (not shown) for yaw control and a pair of horizontal stabilizers **134** each having an elevator flap **136** for pitch control. The fuselage **112** additionally includes an outer surface or skin **138**. It should be appreciated that in other exemplary embodiments of the present disclosure, the aircraft **100** may additionally or alternatively include any other suitable configuration. For example, in other embodiments, the aircraft **100** may include any other control surface configuration.

(50) The exemplary aircraft **100** of FIG. **1** also includes a hybrid-electric propulsion system **150**. For this embodiment, the hybrid-electric propulsion system **150** has a first propulsor **200**A and a second propulsor **200**B both operable to produce thrust. The first propulsor **200**A is mounted to the

first wing **120** and the second propulsor **200**B is mounted to the second wing **122**. Moreover, for the embodiment depicted, the first propulsor **200**A and second propulsor **200**B are each configured in an underwing-mounted configuration. However, in other example embodiments, one or both of the first and second propulsors **200**A, **200**B may be mounted at any other suitable location in other exemplary embodiments.

(51) The first propulsor **200**A includes a gas turbine engine **210**A and one or more electric machines, such as electric machine **300**A mechanically coupled with the gas turbine engine **210**A. The electric machine **300**A can be an electric generator, an electric motor, or a combination generator/motor. For this example embodiment, the electric machine **300**A is a combination generator/motor. In this manner, when operating as an electric generator, the electric machine **300**A can generate electrical power when driven by the gas turbine engine **210**A. When operating as an electric motor, the electric machine **300**A can drive or motor the gas turbine engine **210**A.

(52) Likewise, the second propulsor **200**B includes a gas turbine engine **210**B and one or more electric machines, such as electric machine **300**B mechanically coupled with the gas turbine engine **210**B. The electric machine **300**B can be an electric generator, an electric motor, or a combination generator/motor. For this example embodiment, the electric machine **300**B is a combination generator/motor. In this manner, when operating as an electric generator, the electric machine **300**B can generate electrical power when driven by the gas turbine engine **210**B. When operating as an electric motor, the electric machine **300**B can drive or motor a spool of the gas turbine engine **210**B. Electric machine **300**B can be configured and can operate in a similar manner as electric machine **300**A described herein.

(53) The hybrid-electric propulsion system **150** further includes an electric energy storage unit **180** electrically connectable to the electric machines **300**A, **300**B, and in some embodiments, other electrical loads. In some exemplary embodiments, the electric energy storage unit **180** may include one or more batteries. Additionally, or alternatively, the electric energy storage units **180** may include one or more supercapacitor arrays, one or more ultracapacitor arrays, or both. For the hybrid-electric propulsion system **150** described herein, the electric energy storage unit **180** is configured to store a relatively large amount of electrical power. For example, in certain exemplary embodiments, the electric energy storage unit **180** may be configured to store at least about fifty kilowatt hours of electrical power, such as at least about sixty-five kilowatt hours of electrical power, such as at least about seventy-five kilowatts hours of electrical power, and up to about one thousand kilowatt hours of electrical power.

(54) The hybrid-electric propulsion system **150** also includes a power management system having a controller **182** and a power bus **184**. The electric machines **300**A, **300**B, the electric energy storage unit **180**, and the controller **182** are each electrically connectable to one another through one or more electric lines **186** of the power bus **184**. For instance, the power bus **184** may include various switches or other power electronics movable to selectively electrically connect the various components of the hybrid-electric propulsion system **150**. Particularly, as shown in FIG. **1**, a first power converter **188**A of the power bus **184** is electrically coupled or connectable with the electric machine **300**A via one or more electric lines **186** and a second power converter **188**B of the power bus **184** is electrically coupled or connectable with the electric machine **300**B via one or more electric lines **186**. The power bus **184** may include other power electronics, such as inverters, converters, rectifiers, etc., for conditioning or converting electrical power within the hybrid-electric propulsion system **150**.

(55) The controller **182** is configured to control the power electronics to distribute electrical power between the various components of the hybrid-electric propulsion system **150**. For example, the controller **182** may control the power electronics of the power bus **184** to provide electrical power to, or draw electrical power from, the various components, such as the electric machines **300**A, **300**B, to operate the hybrid-electric propulsion system **150** between various operating modes and perform various functions. Such is depicted schematically as the electric lines **186** of the power bus

184 extend through the controller **182**.
(56) The controller **182** can form a part of a computing system **190** of the aircraft **100**. The computing system **190** of the aircraft **100** can include one or more processors and one or more memory devices embodied in one or more computing devices. For instance, as depicted in FIG. **1**, the computing system **190** includes controller **182** as well as other computing devices, such as computing device **192**. The computing system **190** can include other computing devices as well, such as engine controllers (not shown). The computing devices of the computing system **190** can be communicatively coupled with one another via a communication network. For instance, computing device **192** is located in the cockpit of the aircraft **100** and is communicatively coupled with the controller **182** of the hybrid-electric propulsion system **150** via a communication link **194** of the communication network. The communication link **194** can include one or more wired or wireless communication links.
(57) For this embodiment, the computing device **192** is configured to receive and process inputs, e.g., from a pilot or other crew members, and/or other information. In this manner, as one example, the one or more processors of the computing device **192** can receive an input indicating a command to change a thrust output of the first and/or second propulsors **200**A, **200**B and can cause, in response to the input, the controller **182** to control the electrical power drawn from or delivered to one or both of the electric machines **300**A, **300**B to ultimately change the thrust output of one or both of the propulsors **200**A, **200**B.
(58) The controller **182** and other computing devices of the computing system **190** of the aircraft **100** may be configured in substantially the same manner as the exemplary computing devices of the computing system **950** described below with reference to FIG. **30**.
(59) FIG. **2** provides a schematic view of the first propulsor **200**A of the hybrid-electric propulsion system **150** of the aircraft **100** of FIG. **1**. Although the first propulsor **200**A is shown, it will be appreciated that the second propulsor **200**B can be configured in the same or similar manner as the first propulsor **200**A depicted in FIG. **2**. The exemplary gas turbine engine of FIG. **2** is configured as a single unducted rotor engine **210**A defining an axial direction A, a radial direction R, and a circumferential direction C. The engine **210**A also defines a central longitudinal axis **214**.
(60) As shown in FIG. **2**, the engine **210**A takes the form of an open rotor propulsion system and has a rotor assembly **212** that includes an array of airfoils arranged around the central longitudinal axis **214** of engine **210**A. More particularly, the rotor assembly **212** includes an array of rotor blades **216** arranged around the central longitudinal axis **214** of the engine **210**A. Moreover, as will be explained in more detail below, the engine **210**A also includes a non-rotating vane assembly **218** positioned aft of the rotor assembly **212** (i.e., non-rotating with respect to the central axis **214**). The non-rotating vane assembly **218** includes an array of airfoils also disposed around central axis **214**. More specifically, the vane assembly **218** includes an array of vanes **220** disposed around central longitudinal axis **214**.
(61) The rotor blades **216** are arranged in typically equally-spaced relation around the central longitudinal axis **214**, and each blade has a root **222** and a tip **224** and a span defined therebetween. Similarly, the vanes **220** are also arranged in typically equally-spaced relation around the central longitudinal axis **214**, and each has a root **226** and a tip **228** and a span defined therebetween. The rotor assembly **212** further includes a hub **245** located forward of the plurality of rotor blades **216**.
(62) Additionally, the engine **210**A includes a turbomachine **230** having a core **232** (or high pressure/high speed system) and a low pressure/low speed system. It will be appreciated that as used herein, the terms “speed” and “pressure” are used with respect to the high pressure/high speed system and low pressure/low speed system interchangeably. Further, it will be appreciated that the terms “high” and “low” are used in this same context to distinguish the two systems, and are not meant to imply any absolute speed and/or pressure values.
(63) The core **232** generally includes a high speed compressor **234**, a high speed turbine **236**, and a high speed shaft **238** extending therebetween and connecting the high speed compressor **234** and

high speed turbine **236**. The high speed compressor **234**, the high speed turbine **236**, and the high speed shaft **238** may collectively be referred to as a high speed spool **253** of the engine. Further, a combustion section **240** is located between the high speed compressor **234** and high speed turbine **236**. The combustion section **240** may include one or more configurations for receiving a mixture of fuel and air, and providing a flow of combustion gasses through the high speed turbine **236** for driving the high speed spool **253**.

(64) The low speed system includes a low speed turbine **242**, a low speed compressor **244** or booster, and a low speed shaft **246** extending between and connecting the low speed compressor **244** and low speed turbine **242**. The low speed compressor **244**, the low speed turbine **242**, and the low speed shaft **246** may collectively be referred to as a low speed spool **255** of the engine.

(65) Although the engine **210**A is depicted with the low speed compressor **244** positioned forward of the high speed compressor **234**, in certain embodiments the compressors **234**, **244** may be in an interdigitated arrangement. Additionally, or alternatively, although the engine **210**A is depicted with the high speed turbine **236** positioned forward of the low speed turbine **242**, in certain embodiments the turbines **236**, **242** may similarly be in an interdigitated arrangement.

(66) In order to support the rotating components of the engine **210**A, the engine **210**A includes a plurality of bearings coupling the rotating components to various structural components. Specifically, as depicted in FIG. **2**, bearings **290** support and facilitate rotation of the low speed shaft **246**. Further, bearings **292** support and facilitate rotation of the high speed shaft **238**. Although the bearings **290**, **292** are illustrated as being located generally at forward and aft ends of their associated shafts **246**, **238**, the bearings **290**, **292** may be located at any desired location along their associated shafts. Moreover, in some embodiments, one or more additional bearings other than the bearings **290** shown in FIG. **2** can be used to support the low speed shaft **246**. For instance, in some embodiments, an additional bearing can be positioned at a central or mid-span region of the low speed shaft **246** provides support thereto. Similarly, one or more additional bearings other than the bearings **290** shown in FIG. **2** can be used to support the high-speed shaft **238**. The bearings **290**, **290** can be any suitable type of bearings, such as air bearings, oil-lubricated bearings, etc.

(67) Referring still to FIG. **2**, the turbomachine **230** is generally encased in a cowl **248**. Moreover, it will be appreciated that the cowl **248** defines at least in part an inlet **250** and an exhaust **252**, and includes a turbomachinery flowpath **254** extending between the inlet **250** and the exhaust **252**. The inlet **250** is, for the embodiment shown, an annular or axisymmetric 360 degree inlet **250** located between the rotor assembly **212** and the fixed or stationary vane assembly **218** along the axial direction A, and provides a path for incoming atmospheric air to enter the turbomachinery flowpath **254** (and compressors **244**, **234**, combustion section **240**, and turbines **236**, **242**) inwardly of the guide vanes **220** along the radial direction R. Such a location may be advantageous for a variety of reasons, including management of icing performance as well as protecting the inlet **250** from various objects and materials as may be encountered in operation. In other embodiments, however, the inlet **250** may be positioned at any other suitable location, e.g., aft of the vane assembly **218**, arranged in a non-axisymmetric manner, etc.

(68) As depicted, the rotor assembly **212** is driven by the turbomachine **230**, and more specifically, the low speed spool **255** of the turbomachine **230**. More specifically, for this embodiment, the engine **210**A includes a power gearbox **256**. The rotor assembly **212** is driven by the low speed spool **255** of the turbomachine **230** across the power gearbox **256**. In such a manner, the rotating rotor blades **216** of the rotor assembly **212** may rotate around the central longitudinal axis **214** and generate thrust to propel engine **210**A, and hence, the aircraft **100** (FIG. **1**) to which it is associated, in a forward direction F. The power gearbox **256** can include a gearset for decreasing a rotational speed of the low speed spool **255** relative to the low speed turbine **242** such that the rotor assembly **212** may rotate at a slower rotational speed than the low speed spool **255**.

(69) As briefly noted above, the engine **210**A includes vane assembly **218**. The vane assembly **218**

extends from the cowl **248** and is positioned aft of the rotor assembly **212**. The vanes **220** of the vane assembly **218** may be mounted to a stationary frame or other mounting structure and do not rotate relative to the central longitudinal axis **214**. For reference purposes, FIG. **2** depicts the forward direction with arrow F, which in turn defines the forward and aft portions of the engine **210**A. As shown in FIG. **2**, the rotor assembly **212** is located forward of the turbomachine **230** in a "puller" configuration and the exhaust **252** is located aft of the guide vanes **220**. The vanes **220** of the vane assembly **218** are aerodynamically contoured to straighten out an airflow (e.g., reducing a swirl in the airflow) from the rotor assembly **212** to increase an efficiency of the engine **210**A. For example, the vanes **220** may be sized, shaped, and configured to impart a counteracting swirl to the airflow from the rotor blades **216** so that in a downstream direction aft of both rows of airfoils (e.g., blades **216**, vanes **220**) the airflow has a greatly reduced degree of swirl, which may translate to an increased level of induced efficiency.

(70) In some embodiments, it may be desirable that the rotor blades **216**, the vanes **220**, or both, incorporate a pitch change mechanism such that the airfoils (e.g., blades **216**, vanes **220**, etc.) can be rotated with respect to an axis of pitch rotation either independently or in conjunction with one another. Such pitch change can be utilized to vary thrust and/or swirl effects under various operating conditions, including to adjust a magnitude or direction of thrust produced at the rotor blades **216**, or to provide a thrust reversing feature which may be useful in certain operating conditions, such as upon landing an aircraft, or to desirably adjust acoustic noise produced at least in part by the rotor blades **216**, the vanes **220**, or aerodynamic interactions from the rotor blades **216** relative to the vanes **220**. More specifically, for the embodiment of FIG. **2**, the rotor assembly **212** is depicted with a pitch change mechanism **258** for rotating the rotor blades **216** about their respective pitch axes **260**, and the vane assembly **218** is depicted with a pitch change mechanism **262** for rotating the vanes **220** about their respective pitch axes **264**.

(71) The exemplary single rotor unducted engine **210**A depicted in FIG. **2** is provided by way of example only. Accordingly, it will be appreciated that the engine **210**A may have other suitable configurations. For example, in other example embodiments, the engine **210**A can have other suitable numbers of shafts or spools, turbines, compressors, etc.; fixed-pitch blades or vanes **216**, **220**, or both; a direct-drive configuration (i.e., may not include the gearbox **256**); etc. For example, in other exemplary embodiments, the engine **210**A may be a three-spool engine, having an intermediate speed compressor and/or turbine. In such a configuration, it will be appreciated that the terms "high" and "low," as used herein with respect to the speed and/or pressure of a turbine, compressor, or spool are terms of convenience to differentiate between the components, but do not require any specific relative speeds and/or pressures, and are not exclusive of additional compressors, turbines, and/or spools or shafts.

(72) Additionally or alternatively, in other exemplary embodiments, any other suitable gas turbine engine may be provided. For example, in other exemplary embodiments, the gas turbine engine may be a turboshaft engine, a turboprop engine, turbojet engine, etc. Moreover, for example, although the engine is depicted as a single unducted rotor engine, in other embodiments, the engine may include a multi-stage open rotor configuration, and aspects of the disclosure described hereinbelow may be incorporated therein.

(73) Further, in other exemplary embodiments, the engine **210**A may be configured as a ducted turbofan engine. For example, referring briefly to FIG. **3**, an engine **210**A in accordance with another exemplary embodiment of the present disclosure is depicted. The exemplary embodiment of FIG. **3** may be configured in substantially the same manner as the exemplary engine **210**A described above with respect to FIG. **2** except as noted below. The same or similar reference numerals may refer to the same or similar parts. As shown, the engine **210**A of FIG. **3** includes a nacelle **280** circumferentially surrounding at least in part the rotor assembly **212** and turbomachine **230**, defining a bypass passage **282** therebetween. The vanes **220** of the vane assembly **218** extend between and connect the nacelle **280** with the cowl **248**.

(74) Referring again to FIG. **2**, as noted, the first propulsor **200**A includes electric machine **300**A operably coupled with a rotating component thereof. In this regard, the first propulsor **200**A is an aeronautical hybrid-electric propulsion machine. Particularly, as shown in FIG. **2**, the electric machine **300**A is mechanically coupled with the low speed spool **255** of the gas turbine engine **210**A, and more particularly, the low speed shaft **246** of the low speed spool **255**. As depicted, the electric machine **300**A is embedded within the core of the gas turbine engine **210**A. Specifically, the electric machine **300**A is positioned inward of the turbomachinery flowpath **254** along the radial direction R. Moreover, for this embodiment, the electric machine **300**A is positioned generally at the aft end of the gas turbine engine **210**A and is at least partially overlapping with or aft of the low pressure turbine **242** along the axial direction A. However, in other exemplary embodiments, the electric machine **300**A may be positioned at other suitable locations within the gas turbine engine **210**A. For instance, in some embodiments, the electric machine **300**A can be coupled with the low speed spool **255** in other suitable locations. For instance, in some embodiments, the electric machine **300**A can be positioned forward of the low pressure compressor **244** along the axial direction A and inward of the turbomachinery flowpath **254** along the radial direction R. Further, as shown in FIG. **2**, the electric machine **300**A mechanically coupled with the low speed shaft **246** is electrically coupled with the power bus **184** and is electrically connected to its associated power converter supply **188**A.
(75) In addition or alternatively to the gas turbine engine **210**A having electric machine **300**A coupled to the low speed spool **255**, in some embodiments, the gas turbine engine **210**A can include an electric machine **302**A mechanically coupled with the high speed spool **253** of the gas turbine engine **210**A, and more particularly, the high speed shaft **238** of the high speed spool **253**. As depicted in FIG. **2**, the electric machine **302**A is embedded within the core of the gas turbine engine **210**A and is mechanically coupled with the high speed shaft **238**. The electric machine **302**A is positioned inward of the turbomachinery flowpath **254** along the radial direction R and is positioned forward of the combustion section **140** along the axial direction A. However, in other exemplary embodiments, the electric machine **302**A may be positioned at other suitable locations within the gas turbine engine **210**A. Although not shown, the electric machine **302**A mechanically coupled with the high speed shaft **238** can be electrically coupled with the power bus **184** and can be electrically connected to its own power converter supply.
(76) Like the electric machine **300**A mechanically coupled with the low speed spool **255**, the electric machine **302**A mechanically coupled with the high speed spool **253** can be an electric motor operable to drive or motor the high speed shaft **238**, e.g., during an engine burst. In other embodiments, the electric machine **302**A can be an electric generator operable to convert mechanical energy into electrical energy. In this way, electrical power generated by the electric machine **302**A can be directed to various engine and/or aircraft systems. In some embodiments, the electric machine **302**A can be a motor/generator with dual functionality.
(77) FIG. **4** provides a close-up, schematic view of the electric machine **300**A embedded within the gas turbine engine **210**A. As depicted, the electric machine **300**A defines a centerline **304**, which is aligned with or coaxial with the central longitudinal axis **214** of the gas turbine engine **210**A in this example embodiment. The electric machine **300**A includes a rotor assembly **310** and a stator assembly **340**. The rotor assembly **310** includes a rotor **312** and the stator assembly **340** includes a stator **342**. The rotor **312** of the rotor assembly **310** and the stator **342** of the stator assembly **340** together define an air gap **306** therebetween. Moreover, for this embodiment, the rotor **312** includes a plurality of magnets **314**, such as a plurality of permanent magnets, and the stator **342** includes a plurality of windings or coils **344**. As such, the electric machine **300**A may be referred to as a permanent magnet electric machine. However, in other exemplary embodiments, the electric machine **300**A may be configured in any suitable manner. For example, the electric machine **300**A may be configured as an electromagnetic electric machine, including a plurality of electromagnets and active circuitry, as an induction type electric machine, a switched reluctance type electric

machine, a synchronous AC electric machine, an asynchronous electric machine, or as any other suitable type of electric machine.
(78) The rotor assembly **310** also includes a rotor connection assembly **316**. Generally, the rotor connection assembly **316** operatively couples the rotor **312** with the low speed shaft **246**. As the rotor assembly **310** of the electric machine **300**A is coupled with or attached to the low speed shaft **246**, the rotor assembly **310** is rotatable with the low speed shaft **246**. As shown, the rotor connection assembly **316** has a rotor hub **318** and a rotor connection member **320**. The rotor hub **318** is connected to the low speed shaft **246** and the rotor connection member **320** is connected to the rotor **312**. The rotor hub **318** and the rotor connection member **320** are mechanically coupled or connected with one another at a joint **322**.
(79) For this embodiment, the rotor hub **318** of the rotor connection assembly **316** is connected to the low speed shaft **246** through a splined connection. More particularly, the rotor hub **318** includes a connection portion having a plurality of teeth **324**. Similarly, the low speed shaft **246** includes a connection portion having a plurality of teeth **247**. The plurality of teeth **324** of the rotor hub **318** are configured to engage with the plurality of teeth **247** of the low speed shaft **246**, fixing the two components to one another. In alternative embodiments, the rotor hub **318** may be coupled to the low speed shaft **246** in any other suitable manner.
(80) The stator assembly **340** also includes a stator connection assembly **346**. The stator connection assembly **346** includes a stator connection member **348** that supports the stator **342**. The stator connection member **348** is connected to a structural support member **266** of the turbine section of the gas turbine engine **210**A. The structural support member **266** can be configured as part of an aft frame assembly of the gas turbine engine **210**A. The aft frame assembly can include an aft strut **268** (FIG. **2**) extending through the turbomachinery flowpath **254** along the radial direction R. The aft frame strut **268** provides structural support for the aft end of the cowl **248**.
(81) The gas turbine engine **210**A further includes a cavity wall **270** surrounding at least a portion of the electric machine **300**A. More specifically, the cavity wall **270** substantially completely surrounds the electric machine **300**A, extending from a location proximate a forward end of the electric machine **300**A to a location aft of the electric machine **300**A along the axial direction A. The cavity wall **270** may function as, e.g., a cooling air cavity wall, a sump for cooling fluid, a protective cover for the electric machine **300**A, etc. In some embodiments, the gas turbine engine **210**A may further include a second cavity wall (not shown) to form a buffer cavity surrounding the electric machine **300**A. The buffer cavity formed by the second cavity wall may thermally protect the electric machine **300**A.
(82) During certain operations of the gas turbine engine **210**A, the low speed shaft **246** rotates the rotor assembly **310** of the electric machine **300**A, allowing the electric machine **300**A to generate electrical power. Thus, the electric machine **300**A is operable in a generator mode. In some embodiments, in addition or alternatively to being operable in a generator mode, the electric machine **300**A is operable in a drive mode during certain operations of the gas turbine engine **210**A. In a drive mode, the rotor assembly **310** of the electric machine **300**A drives the low speed shaft **246**. The power converter **188**A (FIG. **2**) can be controlled to provide electrical power to the electric machine **300**A via the electric lines **186**, e.g., when the electric machine **300**A is operating in a drive mode, and electrical power generated by the electric machine **300**A can be carried or transmitted to the power converter **188**A (FIG. **2**) and ultimately to various electrical loads via the electric lines **186**, e.g., when the electric machine **300**A is operating in a generator mode. As shown best in FIG. **2**, the electric lines **186** of the power bus **184** may extend through the turbomachinery flowpath **254** (e.g., through the aft frame strut **268**) and electrically connect the electric machine **300**A to the power converter **188**A and ultimately to one or more electrical loads (accessory systems, electric/hybrid-electric propulsion devices, etc., electrical sources (other electric machines, electric energy storage units, etc., or both.
(83) Although the electric machine **300**A has been described and illustrated in FIG. **4** as having a

particular configuration, it will be appreciated that the inventive aspects of the present disclosure may apply to electric machines having alternative configurations. For instance, the stator assembly **340** and/or rotor assembly **310** may have different configurations or may be arranged in a different manner than illustrated in FIG. **4**. As one example, the electric machine **300**A may have an inner-rotor configuration as shown in FIG. **5** rather than the outer-rotor configuration shown in FIG. **4**. In an inner-rotor configuration, as depicted in FIG. **5**, the rotor **312** is positioned inward of the stator **342** along the radial direction R. In an outer-rotor configuration, as shown in FIG. **4**, the rotor **312** is positioned outward of the stator **342** along the radial direction R. As another example, in some embodiments, the electric machine **300**A may have a tapered configuration in which the rotor **312** and the stator **342** may extend lengthwise along the axial direction A at an angle with respect to the central longitudinal axis **214**, e.g., so that they are not oriented parallel with the central longitudinal axis **214**.

(84) As noted previously, the inventors of the present subject matter have developed architectures and control schemes that may reduce common mode emissions and associated electromagnetic interference in electrical power systems having an electric machine and a power converter system electrically coupled thereto. For instance, in one example aspect, polyphase or multiphase windings of an electric machine are arranged to operate under complementary excitation and PWM excitations are synthesized at the power converter system to reduce common mode emissions by cancelation. Various embodiments of such electrical power systems are provided herein.

(85) With reference now to FIGS. **6**, **7**, and **8**, FIG. **6** provides a schematic view of a power system **400** that includes an electric machine **450** according to one example embodiment of the present disclosure. FIG. **7** provides a schematic view of the multiphase windings of the electric machine **450**. FIG. **8** provides a schematic view of a first power converter unit **418** of a power converter system **410** (FIG. **6**) of the power system **400**. Generally, the power system **400** includes power converter system **410**, electric machine **450**, and a plurality of electrical cables **404**, **405**, **406**, **407** electrically coupling the power converter system **410** and the electric machine **450**. The electric cables **404**, **405**, **406**, **407** can be shielded cables, for example.

(86) For this embodiment, the power converter system **410** is an AC/DC power converter system. For the depicted embodiment, the power converter system **410** has a first power converter unit **418**, a second power converter unit **428**, a third power converter unit **438**, and a fourth power converter unit **448**. The power converter units **418**, **428**, **438**, **448** can be separate or independent units, or alternatively, can be units of a single power converter. In FIG. **6**, the first power converter **418** and the second power converter **428** form a single power converter by connecting the DC output in series or parallel. Similarly, in FIG. **6**, the third power converter **438** and the fourth power converter **448** form a single power converter by connecting the DC output in series or parallel. The first and second power converter units **418**, **428** are associated with a first channel **401** and the third and fourth power converter units **438**, **448** are associated with a second channel **402**. In FIG. **6**, the first channel **401** is a dual DC output channel, and likewise, the second channel **402** is a dual DC output channel. In some alternative embodiments, the first power converter unit **418** can be associated with a first channel, the second power converter unit **428** can be associated with a second channel, the third power converter unit **438** can be associated with a third channel, and the fourth power converter unit **448** can be associated with a fourth channel. In such embodiments, the first, second, third, and fourth channels can be single DC output channels.

(87) The first power converter unit **418** includes first switching elements **411**, the second power converter unit **428** includes second switching elements **412**, the third power converter unit **438** includes third switching elements **413**, and the fourth power converter unit **448** includes fourth switching elements **414**. The first switching elements **411** correspond to all switching elements of the first power converter unit **418**, the second switching elements **412** correspond to all switching elements of the second power converter unit **428**, the third switching elements **413** correspond to all switching elements of the third power converter unit **438**, and the fourth switching elements **414**

correspond to all switching elements of the fourth power converter unit **448**. The first, second, third, and fourth switching elements **411**, **412**, **413**, **414** can be any suitable type of switching elements, such as insulated gate bipolar transistors, power MOSFETs, etc.

(88) The first, second, third, and fourth switching elements **411**, **412**, **413**, **414** can each include switching elements for each phase of power of the power system **400**. By way of example, with reference to FIG. **8**, the first switching elements **411** include a plurality of switching devices or elements associated with each phase. Particularly, for this embodiment, the first switching elements **411** include a first switching element S1, a second switching element S2, a third switching element S3, a fourth switching element S4, a fifth switching element S5, and a sixth switching element S6 associated with an A phase. The first switching elements **411** also include a first switching element S1, a second switching element S2, a third switching element S3, a fourth switching element S4, a fifth switching element S5, and a sixth switching element S6 associated with a B phase. Further, the first switching elements **411** include a first switching element S1, a second switching element S2, a third switching element S3, a fourth switching element S4, a fifth switching element S5, and a sixth switching element S6 associated with a C phase. The second, third, and fourth switching elements **412**, **413**, **414** of the second power converter unit **428**, the third power converter unit **438**, and the fourth power converter unit **448** illustrated schematically in FIG. **6** can be arranged in a similar manner as the first switching elements **411** of the first power converter unit **418** depicted in FIG. **8**.

(89) By turning on or off the switching devices or elements of the first switching elements **411**, the AC phase terminal can be connected to one of the multiple DC bus rails. For the depicted embodiment of FIG. **8**, there exists three DC bus rails, including a positive-DC bus rail **490**, a mid-DC bus rail **492**, and a negative-DC bus rail **494**, constituting a three-level converter unit. Switching elements of a phase leg of the first power converter unit **418** can be connected to more than three DC bus rails in alternative embodiments. In yet other embodiments, a two-level converter unit can also be implemented in which switching elements of a phase leg can be connected to only two DC bus rails, such as a positive-DC bus rail and a negative-DC bus rail. As will be appreciated, turning on or off the switching devices or elements of the second, third, and fourth switching elements **412**, **413**, **414** can connect the AC phase terminal of a given power converter unit **428**, **438**, **448** to one of its multiple DC bus rails in a similar manner as described above with respect to the switching action of the first switching elements **411**.

(90) Moreover, the power system **400** can be grounded by connecting one of the DC bus rails **490**, **492**, **494** to a ground reference, which is typically a conductive frame of the power system **400**. The ground reference can be disconnected or connected to one of the DC bus rails **490**, **492**, **494** through one or more high impedance devices, such as one or more resistors and/or capacitors. There may be parasitic capacitance between a DC rail and the frame inside and outside of the converter, which may provide a path for common mode emissions.

(91) The first, second, third, and fourth switching elements **411**, **412**, **413**, **414** can be controlled by one or more controllable devices. For instance, the switching elements **411**, **412**, **413**, **414** can be controlled by one or more associated gate drivers. For the embodiment depicted in FIG. **6**, one or more first gate drivers **421** are associated with the first switching elements **411**, one or more second gate drivers **422** are associated with the second switching elements **412**, one or more third gate drivers **423** are associated with the third switching elements **413**, and one or more fourth gate drivers **424** are associated with the fourth switching elements **414**. The one or more gate drivers **421**, **422**, **423**, **424** can be controlled to drive or modulate their respective switching elements **411**, **412**, **413**, **414**, e.g., to control the electrical power provided to or drawn from the electric machine **450**.

(92) The first, second, third, and fourth gate drivers **421**, **422**, **423**, **424** can each include one or more gate drivers. For instance, as depicted in FIG. **8**, for the A phase, the first gate drivers **421** include a first driver D1 for driving the first switching element S1, a second driver D2 for driving the second switching element S2, a third driver D3 for driving the third switching element S3, a

fourth driver D4 for driving the fourth switching element S4, a fifth driver D5 for driving the fifth switching element S5, and a second driver D6 for driving the sixth switching element S6. Likewise, for the B phase, the first gate drivers **421** include a first driver D1 for driving the first switching element S1, a second driver D2 for driving the second switching element S2, a third driver D3 for driving the third switching element S3, a fourth driver D4 for driving the fourth switching element S4, a fifth driver D5 for driving the fifth switching element S5, and a second driver D6 for driving the sixth switching element S6. Similarly, for the C phase, the first gate drivers **421** include a first driver D1 for driving the first switching element S1, a second driver D2 for driving the second switching element S2, a third driver D3 for driving the third switching element S3, a fourth driver D4 for driving the fourth switching element S4, a fifth driver D5 for driving the fifth switching element S5, and a second driver D6 for driving the sixth switching element S6. It will be appreciated that, in other example embodiments, more or less than the number of drivers depicted in FIG. **8** are possible. For instance, in some embodiments, one driver can drive multiple switching elements.

(93) The power converter system **410** can also include one or more processors and one or more memory devices. The one or more processors and one or more memory devices can be embodied in one or more controllers or computing devices. For instance, for this embodiment, the one or more processors and one or more memory devices are embodied in a controller **440**. The controller **440** can be communicatively coupled with various devices, such as the gate drivers **421**, **422**, **423**, **424**, one or more sensors, as well as other computing devices. The controller **440** can be communicatively coupled with such devices via a suitable wired and/or wireless connection. Generally, the controller **440** can be configured in the manner illustrated in FIG. **30** and described in the accompanying text.

(94) In alternative embodiments, the one or more processors and one or more memory devices can be embodied in a plurality of controllers. For instance, in some embodiments, each power converter unit **418**, **428**, **438**, **448** can have an associated controller for controlling the switching action of their respective switching elements **411**, **412**, **413**, **414**. In yet other embodiments, the first and second power converter units **418**, **428** associated with the first channel **401** can have a dedicated controller for controlling the switching action of the first and second switching elements **411**, **412**. Likewise, the third and fourth power converter units **438**, **448** associated with the second channel **402** can have a dedicated controller for controlling the switching action of the third and fourth switching elements **413**, **414**.

(95) Furthermore, the power converter system **410** can include one or more electromagnetic interference filters, or EMI filters. For this embodiment, the power converter system **410** includes a first DC-side EMI filter **430** and a first AC-side EMI filter **432** associated with the first channel **401**. The power converter system **410** also includes a second DC-side EMI filter **434** and a second AC-side EMI filter **436** associated with the second channel **402**. Generally, the EMI filters **430**, **432**, **434**, **436** can suppress electromagnetic noise transmitted along their respective channels **401**, **402**. In alternative embodiments, the first channel **401** of the power converter system **410** can include an EMI filter only on the AC side, only on the DC side, or need not include an EMI filter on either the AC or DC side. Additionally or alternatively, the second channel **402** of the power converter system **410** can include an EMI filter only on the AC side, only on the DC side, or need not include an EMI filter on either the AC or DC side.

(96) The electric machine **450** defines an axial direction A (a direction into and out of the page in FIG. **6**), a radial direction R, and a circumferential direction C. The electric machine **450** also defines an axis of rotation AX extending along the axial direction A. Further, as shown, the electric machine **450** has a rotor **452** and a stator **460**. The rotor **452** can be mechanically coupled with a rotating component, such as a rotating component of a gas turbine engine. The rotor **452** is rotatable about the axis of rotation AX. The rotor **452** is depicted outward of the stator **460** along the radial direction R, and thus, the electric machine **450** is configured in an outer-rotor

configuration. However, the inventive aspects of the present disclosure also apply to electric machines having an inner-rotor configuration.

(97) The rotor **452** includes a plurality of magnets **454**. The stator **460** includes a plurality of multiphase windings or coils wound therein, e.g., within slots defined between teeth of the stator **460**. For the embodiment depicted, the stator **460** includes a first multiphase winding **461**, a second multiphase winding **462**, a third multiphase winding **463**, and a fourth multiphase winding **464**.

(98) Each multiphase winding **461**, **462**, **463**, **464** can include windings or coils for various power phases. For instance, as shown best in FIG. **7**, the first multiphase winding **461** includes windings for the first phase A, the second phase B, and the third phase C. For the first phase A of the first multiphase winding **461**, a go-side winding A1 of the first multiphase winding **461** goes through one of the slots of the stator **460**, turns, and a return-side −A2 of the first multiphase winding **461** returns through one of the slots of the stator **460**. For the second phase B of the first multiphase winding **461**, a go-side winding B2 of the first multiphase winding **461** goes through one of the slots of the stator **460**, turns, and a return-side −B1 of the first multiphase winding **461** returns through one of the slots of the stator **460**. For the third phase C of the first multiphase winding **461**, a go-side winding C1 of the first multiphase winding **461** goes through one of the slots of the stator **460**, turns, and a return-side −C2 of the first multiphase winding **461** returns through one of the slots of the stator **460**.

(99) The second multiphase winding **462** also includes windings for the first phase A, the second phase B, and the third phase C. For the first phase A of the second multiphase winding **462**, a go-side winding −A2 of the second multiphase winding **462** goes through one of the slots of the stator **460**, turns, and a return-side A1 of the second multiphase winding **462** returns through one of the slots of the stator **460**. For the second phase B of the second multiphase winding **462**, a go-side winding −B1 of the second multiphase winding **462** goes through one of the slots of the stator **460**, turns, and a return-side B2 of the second multiphase winding **462** returns through one of the slots of the stator **460**. For the third phase C of the second multiphase winding **462**, a go-side winding −C2 of the second multiphase winding **462** goes through one of the slots of the stator **460**, turns, and a return-side C1 of the second multiphase winding **462** returns through one of the slots of the stator **460**.

(100) The third multiphase winding **463** also includes windings for the first phase A, the second phase B, and the third phase C. For the first phase A of the third multiphase winding **463**, a go-side winding −A1 of the third multiphase winding **463** goes through one of the slots of the stator **460**, turns, and a return-side A2 of the third multiphase winding **463** returns through one of the slots of the stator **460**. For the second phase B of the third multiphase winding **463**, a go-side winding −B2 of the third multiphase winding **463** goes through one of the slots of the stator **460**, turns, and a return-side B1 of the third multiphase winding **463** returns through one of the slots of the stator **460**. For the third phase C of the third multiphase winding **463**, a go-side winding −C1 of the third multiphase winding **463** goes through one of the slots of the stator **460**, turns, and a return-side C2 of the third multiphase winding **463** returns through one of the slots of the stator **460**.

(101) The fourth multiphase winding **464** also includes windings for the first phase A, the second phase B, and the third phase C. For the first phase A of the fourth multiphase winding **464**, a go-side winding A2 of the fourth multiphase winding **464** goes through one of the slots of the stator **460**, turns, and a return-side −A1 of the fourth multiphase winding **464** returns through one of the slots of the stator **460**. For the second phase B of the fourth multiphase winding **464**, a go-side winding B1 of the fourth multiphase winding **464** goes through one of the slots of the stator **460**, turns, and a return-side −B2 of the fourth multiphase winding **464** returns through one of the slots of the stator **460**. For the third phase C of the fourth multiphase winding **464**, a go-side winding C2 of the fourth multiphase winding **464** goes through one of the slots of the stator **460**, turns, and a return-side −C1 of the fourth multiphase winding **464** returns through one of the slots of the stator **460**.

(102) As depicted in FIGS. **6** and **8**, the first multiphase winding **461** is electrically coupled with the first switching elements **411** of the power converter system **410**, the second multiphase winding **462** is electrically coupled with the second switching elements **412**, the third multiphase winding **463** is electrically coupled with the third switching elements **413**, and the fourth multiphase winding **464** is electrically coupled with the fourth switching elements **414**. For instance, FIG. **8** depicts the first switching elements **411** being electrically coupled with the A, B, and C phase windings of the first multiphase winding **461**. It will be appreciated that the second, third, and fourth switching elements **412**, **413**, **414** illustrated schematically in FIG. **6** can be electrically coupled with their respective multiphase windings **462**, **463**, **464** in a similar manner as the first switching elements **411** are electrically coupled with the first multiphase winding **461** as depicted in FIG. **8**.

(103) Notably, the first multiphase winding **461** and the second multiphase winding **462** are electrically opposite in phase with respect to one another. That is, the angle of the AC voltage of the second multiphase winding **462** is electrically out of phase with respect to the angle of the AC voltage of the first multiphase winding **461** by one hundred eighty degrees (180°). For instance, for this embodiment, the first multiphase winding **461** has an AC voltage angle of zero degrees (0°) while the second multiphase winding **462** has an AC voltage angle of one hundred eighty degrees (180°). Thus, the first multiphase winding **461** and the second multiphase winding **462** are electrically opposite in phase with respect to one another.

(104) In addition, for this embodiment, the third multiphase winding **463** and the fourth multiphase winding **464** are electrically opposite in phase with respect to one another. Stated another way, the angle of the AC voltage of the fourth multiphase winding **464** is electrically out of phase with respect to the angle of the AC voltage of the third multiphase winding **463** by one hundred eighty degrees (180°). For instance, for this embodiment, the third multiphase winding **463** has an AC voltage angle of zero degrees (0°) while the fourth multiphase winding **464** has an AC voltage angle of one hundred eighty degrees (180°). Accordingly, the third multiphase winding **463** and the fourth multiphase winding **464** are electrically opposite in phase with respect to one another.

(105) As will be explained in greater detail herein, the first and second multiphase windings **461**, **462** associated with the first channel **401** are arranged and configured to operate in opposite angle of AC voltage with respect to one another so that common mode voltage can be canceled or reduced via the PWM control scheme disclosed herein. Particularly, such an arrangement of the first and second multiphase windings **461**, **462** ensures that their common mode voltage waveforms or waveform shapes are the same or nearly the same. Similarly, the third and fourth multiphase windings **463**, **464** associated with the second channel **402** are arranged and configured to operate in opposite angle of AC voltage with respect to one another so that common mode voltage can be canceled or reduced via the PWM control scheme disclosed herein. Particularly, such an arrangement of the third and fourth multiphase windings **463**, **464** ensures that their common mode voltage waveforms or waveform shapes are the same or nearly the same.

(106) As best shown in FIG. **7**, the multiphase windings **461**, **462**, **463**, **464** can be strategically arranged in a two contra-phase winding pair arrangement. Such an arrangement may facilitate balance of radial forces during operation of the electric machine **450**, among other benefits. As shown, the first multiphase winding **461** is arranged opposite the second multiphase winding **462** along the radial direction R and the third multiphase winding **463** is arranged opposite the fourth multiphase winding **464** along the radial direction R. In this regard, the first multiphase winding **461** and the second multiphase winding **462** balance out the radial forces therebetween and the third multiphase winding **463** and the fourth multiphase winding **464** balance out the radial forces therebetween. This allows the multiphase windings **461**, **462** associated with the first channel **401** to be balanced and the multiphase windings **463**, **464** associated with the second channel **402** to be balanced. Further, such a balanced arrangement of the multiphase windings **461**, **462**, **463**, **464** allows for one of the channels to continue operating at its full power even in the event of non-use

or failure of the channel.

(107) Moreover, the stator **460** of the electric machine **450** defines a plurality of sectors or sections. Particularly, for this embodiment, the stator **460** defines four sectors, including a first sector **471**, a second sector **472**, a third sector **473**, and a fourth sector **474**. The sectors **471**, **472**, **473**, **474** are or about of equal size. As shown, the first, second, third, and fourth multiphase windings **461**, **462**, **463**, **464** are wound within a respective one of the sectors of the stator **460**. More specifically, the first multiphase winding **461** is wound within the first sector **471**, the second multiphase winding **462** is wound within the second sector **472**, the third multiphase winding **463** is wound within the third sector **473**, and the fourth multiphase winding **464** is wound within the fourth sector **474**.

(108) With reference now to FIG. **9** in addition to FIGS. **6**, **7**, and **8**, an example control scheme in which common mode emissions associated with the power system **400** can be canceled or reduced will now be provided. FIG. **9** provides a topology of a control system **420** of the power system **400** according to one example embodiment of the present disclosure. As shown, the controller **440**, or more specifically one or more processors thereof, receive one or more system demands **480**. The system demands **480** can be received from a system-level controller or computing device, such as controller **182** (FIG. **1**). The system demands **480** can indicate what is required of the power system **400**, such as a demanded electrical power output or thrust output of the propulsor to which the electric machine **450** is mechanically coupled.

(109) A command generator **446**, which can be a set of computer-executable instructions or logic, can be executed by the one or more processors of the controller **440** to generate one or more commands based at least in part on the system demands **480**. The one or more processors of the controller **440** can execute the command generator **446** to generate voltage, electric current, torque, and/or other demands based at least in part on the system demands **480**. Particularly, as shown in FIG. **9**, the one or more processors of the controller **440** can execute the command generator **446** to generate voltage commands for each multiphase winding **461**, **462**, **463**, **464**. For instance, voltage commands Va1, Vb1, Vc1 can be generated for the first, second, and third phases A, B, C of the first multiphase winding **461**. Similarly, voltage commands Va2, Vb2, Vc2 can be generated for the first, second, and third phases A, B, C of the second multiphase winding **462**. Likewise, voltage commands Va3, Vb3, Vc3 can be generated for the first, second, and third phases A, B, C of the third multiphase winding **463**. Moreover, voltage commands Va4, Vb4, Vc4 can be generated for the first, second, and third phases A, B, C of the fourth multiphase winding **464**.

(110) As further shown in FIG. **9**, the voltage commands generated for each of the multiphase windings **461**, **462**, **463**, **464** can be received by the one or more processors of the controller **440** and can be input into a pulse width modulator **448**. The pulse width modulator **448**, which can be a set of computer-executable instructions or logic, can be executed by the one or more processors of the controller **440** to generate one or more control signals. The one or more processors of the controller **440** can control the switching elements **411**, **412**, **413**, **414** of the power converter system **410** to generate PWM signals based at least in part on the one or more control signals.

(111) Particularly, as shown, the pulse width modulator **448** can include a first modulator **441** for generating first control signals CS1 associated with the first multiphase winding **461**, a second modulator **442** for generating second control signals CS2 associated with the second multiphase winding **462**, a third modulator **443** for generating third control signals CS3 associated with the third multiphase winding **463**, and a fourth modulator **444** for generating fourth control signals CS4 associated with the fourth multiphase winding **464**. The generated control signals CS1 and CS2 can be routed to their respective gate drivers **421**, **422** as shown in FIG. **9**. Based on the received control signals CS1, the gate drivers **421** can drive the first switching elements **411** to generate first PWM signals so as to render a first common mode signal Vcm1. Based on the received control signals CS2, the gate drivers **422** can drive the second switching elements **412** to generate second PWM signals so as to render a second common mode signal Vcm2. Based on the received control signals CS3, the gate drivers **423** can drive the third switching elements **413** to generate third PWM

signals so as to render a third common mode signal Vcm3. Based on the received control signals CS4, the gate drivers **424** can drive the fourth switching elements **414** to generate fourth PWM signals so as to render a fourth common mode signal Vcm4.

(112) The control signals can be generated based at least in part on their associated voltage commands. By way of example, the graph of FIG. **10**A depicts the voltage commands Va1, Vb1, Vc1 associated with the first, second, and third phases A, B, C of the first multiphase winding **461**. In the graph of FIG. **10**A, the amplitude of the voltage commands Va1, Vb1, Vc1 are shown as a function of time. Upon executing the first modulator **441** of the pulse width modulator **448**, the one or more processors of the controller **440** can generate first control signals CS1. For instance, a first control signal CS1 associated with an A phase of the power system **400** can be generated based at least in part on the voltage command Va1. The first switching elements **411** can be controlled based on the first control signal CS1 associated with the A phase to generate a first PWM signal associated with the A phase of the power system **400**, denoted as PWM-Va1 in the graph of FIG. **10**B.

(113) Likewise, upon executing the first modulator **441** of the pulse width modulator **448**, the one or more processors of the controller **440** can generate a first control signal CS1 associated with a B phase of the power system **400** based at least in part on the voltage command Vb1. Although not depicted in FIGS. **10**A through **10**D (see FIG. **9**), a first PWM signal PWM-vb1 associated with the B phase of the power system **400** can be generated based at least in part on the first control signal CS1 associated with the B phase. Similarly, upon executing the first modulator **441** of the pulse width modulator **448**, the one or more processors of the controller **440** can generate a first control signal CS1 associated with a C phase of the power system **400** based at least in part on the voltage command Vc1. Although not depicted in FIGS. **10**A through **10**D (see FIG. **9**), a first PWM signal PWM-vc1 associated with the C phase of the power system **400** can be generated based at least in part on the first control signal CS1 associated with the C phase. Accordingly, first PWM signals PWM-Va1, PWM-Vb1, PWM-Vc1 can be generated based at least in part on their respective voltage commands Va1, Vb1, Vc1. The graph of FIG. **11**A depicts the voltage commands Va1, Vb1, Vc1 associated with the first, second, and third phases A, B, C of the first multiphase winding **461** (as does the graph of FIG. **10**A). In the graph of FIG. **11**A, the amplitude of the voltage commands Va1, Vb1, Vc1 are shown as a function of time.

(114) The first control signals/PWM signals can be generated using any suitable technique. As one example technique, the voltage command signals can be compared against or to a carrier signal. For instance, as shown in the graph of FIG. **11**B, the voltage command Va1 can be compared against or to a carrier signal CS-V1. For this embodiment, the carrier signal CS-V1 has a triangular-shaped pattern, however, the carrier signal CS-V1 may have other suitable patterns or waveforms in other embodiments. In comparing the voltage command signal Va1 to the carrier signal CS-V1, in instances in which the voltage command Va1 is above the carrier signal CS-V1, the first switching elements **411** are controlled so that the generated first PWM signal PWM-Va1 has a maximum amplitude at that instance. In contrast, in instances in which the voltage command Va1 is below the first carrier signal CS-V1, the first switching elements **411** are controlled so that the generated first PWM signal PWM-Va1 has a minimum amplitude at that instance. In this manner, the first PWM signal Va1 can be generated based at least in part on its associated voltage command Va1. The first PWM signal PWM-Vb1 associated with voltage command Vb1 and the first PWM signal PWM-Vc1 associated with the voltage command Vc1 can be generated using the carrier-signal CS-V1 in the same manner. As a result, the first PWM signals can be generated based on their respective voltage commands using the comparison technique.

(115) Generation of the first PWM signals PWM-Va1, PWM-Vb1, PWM-Vc1 renders a first common mode voltage signal Vcm1. The rendered first common mode voltage signal Vcm1 is shown depicted in the graph of FIG. **10**C. The pattern or waveform of the first common mode voltage signal Vcm1 takes into account the first PWM signal PWM-Va1 generated based on

voltage command Va1, the first PWM signal PWM-Vb1 generated based on voltage command Vb1, and the first PWM signal PWM-Vc1 generated based on voltage command Vc1. Particularly, for a given instance in time, the amplitude of the first common mode voltage signal Vcm1 can be an average amplitude of the three generated first PWM signals PWM-Va1, PWM-Vb1, PWM-Vc1 associated with the first multiphase winding **461**. Stated another way, the first common mode voltage signal Vcm1 can be determined according to the following equation:

(116) $V_{cm1} = \frac{1}{3}(\text{PWM} - Va1 + \text{PWM} - Vb1 + \text{PWM} - Vc1)$ (Equation1)

wherein V.sub.cm1 is the amplitude of the first common mode voltage signal, PWM-Va1 is the pulse width modulation signal generated based on voltage command Va1, PWM-Vb1 is the pulse width modulation signal generated based on voltage command Vb1, and PWM-Vc1 is the pulse width modulation signal generated based on voltage command Vc1.

(117) Generation of second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 renders the second common mode voltage signal Vcm2. The rendered second common mode voltage signal Vcm2 associated with the second multiphase winding **462** is depicted in the graph of FIG. **10**D. The second common mode voltage signal Vcm2 can be rendered in a similar manner as the first common mode voltage signal Vcm1, except as provided below.

(118) The graph of FIG. **12**A depicts the voltage commands Va2, Vb2, Vc2 associated with the first, second, and third phases A, B, C of the second multiphase winding **462**. In the graph of FIG. **12**A, the amplitude of the voltage commands Va2, Vb2, Vc2 are shown as a function of time. Notably, as the second multiphase winding **462** is arranged and wound to operate electrically opposite in phase with respect to the first multiphase winding **461**, voltage commands Va2, Vb2, Vc2 are shown having opposite waveforms with respect to voltage commands Va1, Vb1, Vc1 (see the graph of FIG. **11**B).

(119) Upon executing the second modulator **442** of the pulse width modulator **448**, the one or more processors of the controller **440** can generate second control signals CS2. For instance, a second control signal CS2 associated with the A phase of the power system **400** can be generated based at least in part on the voltage command Va2. The second switching elements **412** can be controlled based on the second control signal CS2 associated with the A phase to generate a second PWM signal PWM-Va2 associated with the A phase of the power system **400**. A second control signal CS2 associated with the B phase of the power system **400** can be generated based at least in part on the voltage command Vb2. The second switching elements **412** can be controlled based on the second control signal CS2 associated with the B phase to generate a second PWM signal PWM-Vb2 associated with the B phase of the power system **400**. Further, a second control signal CS2 associated with the C phase of the power system **400** can be generated based at least in part on the voltage command Vc2. The second switching elements **412** can be controlled based on the second control signal CS2 associated with the C phase to generate a second PWM signal PWM-Vc2 associated with the C phase of the power system **400**.

(120) The second control signals/PWM signals can be generated using any suitable technique. As noted above, in some example embodiments a voltage command signal can be compared against or to a carrier signal. For instance, as shown in the graph of FIG. **12**B, the voltage command signal Va2 can be compared against or to a second carrier signal CS-V2. For this embodiment, the second carrier signal CS-V2 has a triangular-shaped pattern, however, the second carrier signal CS-V2 may have other suitable patterns or waveforms in other embodiments. Notably, the one or more processors of the controller **440** have shifted the second carrier signal CS-V2 by one hundred eighty degrees (180°) with respect to the first carrier signal CS-V1 (see the graph of FIG. **11**B). In this regard, the second carrier signal CS-V2 is an inverse of the first carrier signal CS-V1.

(121) Referring to the graph of FIG. **12**B, in comparing the voltage command Va2 to the second carrier signal CS-V2, in instances in which the voltage command Va2 is above the second carrier signal CS-V2, the second switching elements **412** are controlled so that the generated second PWM

signal PWM-Va2 has a maximum amplitude at that instance. In contrast, in instances in which the voltage command Va2 is below the second carrier signal CS-V2, the second switching elements **412** are controlled so that the generated second PWM signal PWM-Va2 has a minimum amplitude at that instance. In this manner, the second PWM signal PWM-Va2 can be generated based at least in part on its associated voltage command Va2. The second PWM signals PWM-Vb2, PWM-Vc2 can be generated using the second carrier signal CS-V2 and their respective voltage commands Vb2, Vc2 in the same manner.

(122) The pattern or waveform of the rendered second common mode voltage signal Vcm2 takes into account the second PWM signal PWM-Va2 generated based on voltage command Va2, the second PWM signal PWM-Vb2 generated based on voltage command Vb2, and the second PWM signal PWM-Vc2 generated based on voltage command Vc2. Particularly, for a given instance in time, the amplitude of the second common mode voltage signal Vcm2 can be an average amplitude of the three generated second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 associated with the second multiphase winding **462**. Stated differently, the second common mode voltage signal Vcm2 can be determined according to the following equation:

(123) $V_{cm2} = \frac{1}{3}(\text{PWM} - Va2 + \text{PWM} - Vb2 + \text{PWM} - Vc2)$ (Equation2)

wherein V.sub.cm2 is the amplitude of the second common mode voltage signal, PWM-Va2 is the pulse width modulation signal generated based on voltage command Va2, PWM-Vb2 is the pulse width modulation signal generated based on voltage command Vb2, and PWM-Vc2 is the pulse width modulation signal generated based on voltage command Vc2.

(124) As will be appreciated by comparing the rendered first common mode signal Vcm1 depicted in the graph of FIG. **10**C with the rendered second common mode signal Vcm2 depicted in the graph of FIG. **10**D, the second common mode signal Vcm2 has the same or nearly the same waveform as the first common mode signal Vcm1 since the first multiphase winding **461** and the second multiphase winding **462** are arranged and configured to operate electrically opposite in phase with respect to one another. Moreover, shifting the second carrier signal CS-V2 by one hundred eighty degrees (180°) with respect to the first carrier signal CS-V1 effectively shifts the polarity of the second common mode signal Vcm2 with respect to the first common mode signal Vcm1.

(125) Accordingly, when the first switching elements **411** are modulated by their associated gate drivers **421** based at least in part on the first control signals CS1, the first PWM signals PWM-Va1, PWM-Vb1, PWM-Vc1 are generated so as to render the first common mode signal Vcm1. Likewise, when the second switching elements **412** are modulated by their associated gate drivers **422** based at least in part on the second control signals CS2, the second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 are generated so as to render the second common mode signal Vcm2. As the second common mode signal Vcm2 has the same or nearly the same waveform as the first common mode signal Vcm1 but with opposite polarity, common mode emissions associated with the first channel **401** can be canceled or reduced. Thus, a number of benefits may be realized. For instance, the need for EMI filters can be eliminated or at least one or more of the EMI filters can be reduced in size. Moreover, cancelation or reduction of the common mode emissions can reduce shaft voltage and bearing currents, thereby potentially: reducing bearing stress, eliminating the need for a shaft grounding brush, eliminating the need of a bearing insulation sleeve or ceramic bearing, and/or reducing leakage current through shaft loads, such as gears or sensors. Other benefits and advantages may be realized as well. Notably, cancellation of common mode emissions can be achieved in embodiments in which the first power converter unit **418** and the second power converter unit **428** are connected in series, parallel, or are independent units.

(126) FIG. **13** depicts a simplified graph depicting various signals as a function of time. Specifically, FIG. **13** shows the generated first PWM signals PWM-Va1, PWM-Vb1, PWM-Vc1 and the rendered first common mode signal Vcm1 as well as the generated second PWM signals

PWM-Va2, PWM-Vb2, PWM-Vc2 and the rendered second common mode signal Vcm2. As shown, the first common mode signal Vcm1 “mirrors” the second common mode signal Vcm2. Consequently, the sum of the first common mode signal Vcm1 and the second common mode signal Vcm2 is essentially a flat constant line, which is representative of ideal common mode cancelation.

(127) As will be appreciated, common mode emissions associated with the second channel **402** can be reduced or eliminated in the same or similar manner as described above with respect to the first channel **401**. Generally, a third common mode voltage signal Vcm3 can be rendered in the same or similar as described above with respect to the first common mode signal Vcm1. Particularly, upon executing the third modulator **443** of the pulse width modulator **448**, the one or more processors of the controller **440** can generate third control signals CS3. For instance, a third control signal CS3 associated with the A phase of the power system **400** can be generated based at least in part on voltage command Va3, a third control signal CS3 associated with the B phase of the power system **400** can be generated based at least in part on voltage command Vb3, and a third control signal CS3 associated with the C phase of the power system **400** can be generated based at least in part on voltage command Vc3. The third switching elements **413** can be controlled based on the third control signals CS3 to generate third PWM signals PWM-Va3, PWM-Vb3, PWM-Vc3 so as to render third common mode voltage signal Vcm3.

(128) Moreover, generally, a fourth common mode voltage signal Vcm4 can be rendered in the same or similar as described above with respect to the second common mode signal Vcm2. Particularly, upon executing the fourth modulator **444** of the pulse width modulator **448**, the one or more processors of the controller **440** can generate fourth control signals CS4. For instance, a fourth control signal CS4 associated with the A phase of the power system **400** can be generated based at least in part on voltage command Va4, a fourth control signal CS4 associated with the B phase of the power system **400** can be generated based at least in part on voltage command Vb4, and a fourth control signal CS4 associated with the C phase of the power system **400** can be generated based at least in part on voltage command Vc4. The fourth switching elements **414** can be controlled based on the fourth control signals CS4 to generate fourth PWM signals PWM-Va4, PWM-Vb4, PWM-Vc4 so as to render fourth common mode voltage signal Vcm4.

(129) Notably, in rendering the third and fourth common mode voltage signals Vcm3, Vcm4, the fourth common mode voltage Vcm4 has the same or nearly the same waveform as the third common mode voltage Vcm3 since the third multiphase winding **463** and the fourth multiphase winding **464** are arranged and configured to operate electrically opposite in phase with respect to one another. Moreover, shifting the carrier signal associated with the fourth multiphase winding **464** by one hundred eighty degrees (180°) with respect to the carrier signal associated with the third multiphase winding **463** shifts the polarity of the fourth common mode voltage signal Vcm4 with respect to the third common mode voltage signal Vcm3. Thus, the third common mode signal Vcm3 and the fourth common mode signal Vcm4 have the same or nearly the same waveform with opposite polarity. As the fourth common mode signal Vcm4 has the same or nearly the same waveform as the third common mode signal Vcm3 but with opposite polarity, common mode emissions associated with the second channel **402** can be canceled or reduced. In this way, the benefits noted above may be realized.

(130) It will be appreciated that the topology of the control system **420** is a non-limiting example. For instance, in some embodiments, the control system **420** can include a plurality of controllers instead of a single controller. In such embodiments, each controller can include its own command generator or set of computer-executable instructions or logic and PW modulator. For instance, in some example embodiments, a first controller associated with the first channel **401** can include a command generator that, when executed by the one or more processors of the first controller, cause the one or more processors of the first controller to generate one or more commands associated with the first and second multiphase windings **461**, **462** based at least in part on the system

demands **480**. The first controller can further include first modulator **441** and second modulator **442** for generating first control signals CS1 associated with the first multiphase winding **461** and for generating second control signals CS2 associated with the second multiphase winding **462**, respectively. Further, a second controller associated with the second channel **402** can include a command generator that, when executed by the one or more processors of the second controller, cause the one or more processors of the second controller to generate one or more commands associated with the third and fourth multiphase windings **463**, **464** based at least in part on the system demands **480**. The second controller can further include third modulator **443** and fourth modulator **444** for generating third control signals CS3 associated with the third multiphase winding **463** and for generating fourth control signals CS4 associated with the fourth multiphase winding **464**, respectively. Other topologies are possible. For instance, the control system **420** can include at least one controller dedicated to controlling each power converter unit.

(131) The inventive aspects of the present disclosure may apply to other multichannel power systems as well. Particularly, in some embodiments, common mode emissions can be canceled or reduced between two power channels instead of within a single power channel as disclosed above with respect to the multichannel embodiment of FIG. **6**.

(132) By way of example, FIG. **14** provides a schematic view of an electrical power system **500** according to one example embodiment of the present disclosure. As will be explained below, the power system **500** of FIG. **14** is arranged and configured to operate so that common mode emissions can be canceled or reduced between its two power channels. As depicted, generally, the power system **500** of FIG. **14** includes power converter system **510**, electric machine **550**, and one or more electrical cables **504**, **506** electrically coupling the power converter system **510** and the electric machine **550**. The electric cables **504**, **506** can be shielded cables, for example.

(133) For the depicted embodiment of FIG. **14**, the power converter system **510** is an AC/DC power converter system. As shown in FIG. **14**, the power converter system **510** has a first power converter unit **518** and a second power converter unit **528**. The power converter units **518**, **528** can be separate or independent units, or alternatively, can be units of a single power converter. In FIG. **14**, the first power converter unit **518** and the second power converter unit **528** are independent units. In alternative embodiments, the first power converter unit **518** and the second power converter unit **528** can form a single power converter by connecting the first power converter unit **518** and the second power converter unit **528** in series or parallel. The first power converter unit **518** includes first switching elements **511** and the second power converter unit **528** includes second switching elements **512**. The first power converter unit **518** is associated with a first channel **501** and the second power converter unit **528** is associated with a second channel **502**. The first and second switching elements **511**, **512** can be any suitable type of switching elements, such as insulated gate bipolar transistors, power MOSFETs, etc.

(134) The first and second switching elements **511**, **512** can each include switching elements for each phase of power of the power system **500**. For instance, the first switching elements **511** can include switching elements associated with an A phase, switching elements associated with a B phase, and switching elements associated with a C phase of the power system **500**. Likewise, the second switching elements **512** can include switching elements associated with the A phase, switching elements associated with the B phase, and switching elements associated with the C phase of the power system **500**.

(135) The first and second switching elements **511**, **512** can be controlled by one or more controllable devices. For instance, the first and second switching elements **511**, **512** can be controlled by one or more associated gate drivers. For the embodiment depicted in FIG. **14**, one or more first gate drivers **521** are associated with the first switching elements **511** and one or more second gate drivers **522** are associated with the second switching elements **512**. The one or more gate drivers **521**, **522** can be controlled to drive or modulate their respective switching elements **511**, **512**, e.g., to control the electrical power provided to or drawn from the electric machine **550**.

The first and second gate drivers **521**, **522** can each include one or more gate drivers.
(136) The power converter system **510** can also include one or more processors and one or more memory devices. The one or more processors and one or more memory devices can be embodied in one or more controllers or computing devices. For instance, for this embodiment, the one or more processors and one or more memory devices are embodied in a controller **540**. The controller **540** can be communicatively coupled with various devices, such as the gate drivers **521**, **522**, one or more sensors (e.g., current and/or voltage sensors), as well as other computing devices. The controller **540** can be communicatively coupled with such devices via a suitable wired and/or wireless connection. Generally, the controller **540** can be configured and perform operations in the manner illustrated in FIG. **30** and described in the accompanying text.
(137) Furthermore, the power converter system **510** can include one or more EMI filters. For this embodiment, the power converter system **510** includes a DC-side EMI filter **530** and an AC-side EMI filter **532**. Generally, the EMI filters **530**, **532** can suppress electromagnetic noise transmitted along the first and second channels **501**, **502**.
(138) The electric machine **550** defines an axial direction A (a direction into and out of the page in FIG. **14**), a radial direction R, and a circumferential direction C. The electric machine **550** also defines an axis of rotation AX extending along the axial direction A. Further, as shown, the electric machine **550** has a rotor **552** and a stator **560**. The rotor **552** can be mechanically coupled with a rotating component, such as a rotating component of a gas turbine engine. The rotor **552** is rotatable about the axis of rotation AX. The rotor **552** is depicted outward of the stator **560** along the radial direction R, and thus, the electric machine **550** is configured in an outer-rotor configuration. However, the inventive aspects of the present disclosure also apply to electric machines having an inner-rotor configuration, as previously noted.
(139) The rotor **552** includes a plurality of magnets **554**. The stator **560** includes a plurality of multiphase windings or coils wound therein, e.g., within slots defined between teeth of the stator **560**. For the embodiment depicted, the stator **560** includes a first multiphase winding **561** and a second multiphase winding **562**. The first multiphase winding **561** is associated with the first channel **501** while the second multiphase winding **562** is associated with the second channel **502**. Each multiphase winding **561**, **562** can include windings or coils for various power phases. For instance, the first multiphase winding **561** can include windings for the first phase A, the second phase B, and the third phase C. Likewise, the second multiphase winding **562** can include windings for the first phase A, the second phase B, and the third phase C. Moreover, as depicted schematically in FIG. **14**, the first multiphase winding **561** is electrically coupled with the first switching elements **511** of the power converter system **510** and the second multiphase winding **562** is electrically coupled with the second switching elements **512**.
(140) Notably, the first multiphase winding **561** associated with the first channel **501** and the second multiphase winding **562** associated with the second channel **502** are electrically opposite in phase with respect to one another. That is, the angle of the AC voltage of the second multiphase winding **562** is electrically out of phase with respect to the angle of the AC voltage of the first multiphase winding **561** by one hundred eighty degrees (180°). For instance, for this embodiment, the first multiphase winding **561** has an AC voltage angle of zero degrees (0°) while the second multiphase winding **562** has an AC voltage angle of one hundred eighty degrees (180°). Thus, the first multiphase winding **561** and the second multiphase winding **562** are electrically opposite in phase with respect to one another. As will be explained below, the first multiphase winding **561** associated with the first channel **501** and second multiphase winding **562** associated with the second channel **502** are arranged and configured to operate in opposite angle of AC voltage with respect to one another. This arrangement, along with the PWM control scheme disclosed herein, facilitates cancelation of common mode emissions.
(141) Further, as best shown in FIG. **14**, the first and second multiphase windings **561**, **562** can be strategically arranged in an interweaved contra-phase winding pair arrangement. Such an

arrangement may facilitate balance of radial forces during operation of the electric machine **550**, among other benefits. As shown, the first multiphase winding **561** is arranged in a first set **565** and a second set **566**. The first set **565** and the second set **566** of the first multiphase winding **561** are positioned opposite one another along the radial direction R. In this regard, when the first channel **501** is in operation, the first set **565** and the second set **566** of the first multiphase winding **561** balance out the radial forces therebetween. Such a balanced arrangement of the first multiphase winding **561** allows for the first channel **501** to continue operating at its full power even in the event of non-use or failure of the second channel **502**. The first set **565** and the second set **566** can be electrically connected to one another in series or parallel.

(142) The second multiphase winding **562** is also arranged in a first set **567** and a second set **568**. The first set **567** and the second set **568** of the second multiphase winding **562** are positioned opposite one another along the radial direction R. The first set **567** and the second set **568** can be electrically connected to one another in series or parallel. As depicted, the first set **565** of the first multiphase winding **561** is positioned between the first set **567** and the second set **568** of the second multiphase winding **562** along the circumferential direction C and the second set **566** of the first multiphase winding **561** is positioned between the first set **567** and the second set **568** of the second multiphase winding **562** along the circumferential direction C. In this way, when the second channel **502** is in operation, the first set **567** and the second set **568** of the second multiphase winding **562** balance out the radial forces therebetween. Such a balanced arrangement of the second multiphase winding **562** allows for the second channel **502** to continue operating at its full power even in the event of non-use or failure of the first channel **501**.

(143) Moreover, the stator **560** of the electric machine **550** defines a plurality of sectors or sections. Particularly, for this embodiment, the stator **560** defines four sectors, including a first sector **571**, a second sector **572**, a third sector **573**, and a fourth sector **574**. The sectors **571**, **572**, **573**, **574** are or about of equal size. As shown, the first and second sets **565**, **566** of the first multiphase winding **561** and the first and second sets **567**, **568** of the second multiphase winding **562** are each wound within a respective one of the sectors **571**, **572**, **573**, **574** of the stator **560**. More specifically, the first set **565** of the first multiphase winding **561** is wound within the first sector **571**, the second set **566** of the first multiphase winding **561** is wound within the second sector **572**, the first set **567** of the second multiphase winding **562** is wound within the third sector **573**, and the second set **568** of the second multiphase winding **562** is wound within the fourth sector **574**.

(144) With reference now to FIG. **15** in addition to FIG. **14**, an example control scheme in which common mode emissions associated with the power system **500** can be canceled or reduced will now be provided. FIG. **15** provides a topology of a control system **520** of the power system **500**. It will be appreciated that the topology of the control system **520** is intended to be a non-limiting example. For instance, in other embodiments, the control system **520** can include a plurality of controllers, e.g., one for controlling the first switching elements **511** and one for controlling the second switching elements **512**.

(145) As shown in FIG. **15**, the controller **540**, or more specifically one or more processors thereof, receive one or more system demands **580**. The system demands **580** can be received from a system-level controller or computing device, such as controller **182** (FIG. **1**). The system demands **580** can indicate what is required of the power system **500**, such as a demanded electrical power output or thrust output of the propulsor to which the electric machine **550** is mechanically coupled.

(146) A command generator **546**, which can be a set of computer-executable instructions or logic, can be executed by the one or more processors of the controller **540** to generate one or more commands based at least in part on the system demands **580**. For example, the one or more processors of the controller **540** can execute the command generator **546** to generate voltage, electric current, torque, and/or other demands based at least in part on the system demands **580**. Particularly, as shown in FIG. **15**, the one or more processors of the controller **540** can execute the command generator **546** to generate voltage commands for each multiphase winding **561**, **562**. For

instance, voltage commands Va1, Vb1, Vc1 can be generated for the first, second, and third phases A, B, C of the first multiphase winding **561**. Similarly, voltage commands Va2, Vb2, Vc2 can be generated for the first, second, and third phases A, B, C of the second multiphase winding **562**.
(147) The voltage commands generated for each of the multiphase windings **561**, **562** can be received by the one or more processors of the controller **540** and can be input into a pulse width modulator **548**. The pulse width modulator **548**, which can be a set of computer-executable instructions or logic, can be executed by the one or more processors of the controller **540** to generate control signals. The one or more processors of the controller **440** can control the switching elements **511**, **512**, of the power converter system **510** to generate PWM signals based at least in part on the one or more control signals.
(148) Particularly, as shown, the pulse width modulator **548** can include a first modulator **541** for generating first control signals CS1 associated with the first multiphase winding **561** and a second modulator **542** for generating second control signals CS2 associated with the second multiphase winding **562**. The generated control signals CS1 and CS2 can be routed to their respective gate drivers **521**, **522** as shown in FIG. **15**. Based on the received control signals CS1, the gate drivers **521** can drive the first switching elements **511** to generate first PWM signals so as to render a first common mode signal Vcm1. Based on the received control signals CS2, the gate drivers **522** can drive the second switching elements **512** to generate second PWM signals so as to render a second common mode signal Vcm2.
(149) The first control signals CS1 can be generated based at least in part on their associated voltage commands. Particularly, upon executing the first modulator **541** of the pulse width modulator **548**, the one or more processors of the controller **540** can generate first control signals CS1. For instance, a first control signal CS1 associated with the A phase of the power system **500** can be generated based at least in part on the voltage command Va1, a first control signal CS1 associated with the B phase of the power system **500** can be generated based at least in part on the voltage command Vb1, and a first control signal CS1 associated with the C phase of the power system **500** can be generated based at least in part on the voltage command Vc1. The first switching elements **511** can be controlled based on the first control signal CS1 associated with the A phase to generate a first PWM signal PWM-Va1 associated with the A phase of the power system **500**. Similarly, the first switching elements **511** can be controlled based on the first control signal CS1 associated with the B phase to generate a first PWM signal PWM-Vb1 associated with the B phase of the power system **500**. Likewise, the first switching elements **511** can be controlled based on the first control signal CS1 associated with the C phase to generate a first PWM signal PWM-Vc1 associated with the C phase of the power system **500**. The first PWM signals PWM-Va1, PWM-Vb1, PWM-Vc1 can be generated using any suitable technique, such as the carrier signal comparison technique disclosed above.
(150) Based on the received control signals CS1, the gate drivers **521** can drive the first switching elements **511** to generate first PWM signals PWM-Va1, PWM-Vb1, PWM-Vc1 so as to render a first common mode signal Vcm1. The pattern or waveform of the first common mode voltage signal Vcm1 takes into account the first PWM signal PWM-Va1 generated based on voltage command Va1, the first PWM signal PWM-Vb1 generated based on voltage command Vb1, and the PWM signal PWM-Vc1 generated based on voltage command Vc1. Particularly, for a given instance, the amplitude of the first common mode voltage signal Vcm1 can be an average amplitude of the three generated PWM signals PWM-Va1, PWM-Vb1, PWM-Vc1 associated with the first multiphase winding **561**.
(151) The second control signals CS2 can be generated based at least in part on their associated voltage commands. Particularly, upon executing the second modulator **542** of the pulse width modulator **548**, the one or more processors of the controller **540** can generate second control signals CS2. For instance, a second control signal CS2 associated with the A phase of the power system **500** can be generated based at least in part on the voltage command Va2, a second control

signal CS2 associated with the B phase of the power system **500** can be generated based at least in part on the voltage command Vb2, and a second control signal CS2 associated with the C phase of the power system **500** can be generated based at least in part on the voltage command Vc2.

(152) The second switching elements **512** can be controlled based on the second control signal CS2 associated with the A phase to generate a second PWM signal PWM-Va2 associated with the A phase of the power system **500**. Similarly, the second switching elements **512** can be controlled based on the second control signal CS2 associated with the B phase to generate a second PWM signal PWM-Vb2 associated with the B phase of the power system **500**. Likewise, the second switching elements **512** can be controlled based on the second control signal CS2 associated with the C phase to generate a second PWM signal PWM-Vc2 associated with the C phase of the power system **500**.

(153) Based on the received control signals CS2, the gate drivers **422** can drive the second switching elements **412** to generate the second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 so as to render the second common mode signal Vcm2. The pattern or waveform of the second common mode voltage signal Vcm2 takes into account the second PWM signal PWM-Va2 generated based on voltage command Va2, the second PWM signal PWM-Vb2 generated based on voltage command Vb2, and the second PWM signal PWM-Vc2 generated based on voltage command Vc2. Particularly, for a given instance, the amplitude of the second common mode voltage signal Vcm2 can be an average amplitude of the three generated PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 associated with the second multiphase winding **562**.

(154) The second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 can be generated using any suitable technique, such as the carrier signal comparison technique disclosed above. Notably, the one or more processors of the controller **540** can shift the polarity of a second carrier signal against which the voltage commands Va2, Vb2, Vc2 are compared (e.g., by one hundred eighty degrees (180°) with respect to a first carrier signal against which the voltage commands Va1, Vb1, Vc1 are compared) so that the rendered second common mode signal Vcm2 is ultimately opposite in polarity with respect to the rendered first common mode signal Vcm1.

(155) Notably, the rendered second common mode signal Vcm2 has the same or nearly the same waveform as the rendered first common mode signal Vcm1 since the first multiphase winding **561** and the second multiphase winding **562** are arranged and configured to operate electrically opposite in phase with respect to one another. Moreover, shifting the second carrier signal associated with the second multiphase winding **562** by one hundred eighty degrees (180°) with respect to first carrier signal associated with the first multiphase winding **561** effectively shifts the polarity of the second common mode voltage signal Vcm2 with respect to the first common mode voltage signal Vcm1.

(156) Accordingly, when the first switching elements **511** are modulated by their associated gate drivers **521** based at least in part on the first control signals CS1, the first PWM signals PWM-Va1, PWM-Vb1, PWM-Vc1 are generated so as to render the first common mode signal Vcm1. Likewise, when the second switching elements **512** are modulated by their associated gate drivers **522** based at least in part on the second control signals CS2, the second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 are generated so as to render the second common mode signal Vcm2. As the second common mode signal Vcm2 has the same or nearly the same waveform as the first common mode signal Vcm1 but with opposite polarity, common mode emissions between the first channel **501** and the second channel **502** can be canceled or reduced as the first common mode voltage signal Vcm1 and the second common mode voltage signal Vcm2 have the same or nearly the same waveform with opposite polarity. Thus, a number of benefits may be realized. For instance, the need for EMI filters can be eliminated or at least one or more of the EMI filters can be reduced in size. Moreover, cancelation or reduction of the common mode emissions can reduce shaft voltage and bearing currents, thereby potentially: reducing bearing stress, eliminating the need for a shaft grounding brush, eliminating the need of a bearing insulation sleeve or ceramic

bearing, and/or reducing leakage current through shaft loads, such as gears or sensors. Other benefits and advantages may be realized as well.
(157) In some embodiments, the first channel **501** and the second channel **502** are paired with the same electrical load. Doing so may keep the operating conditions and load level the same or nearly the same. Advantageously, this may enhance the ability of the system **500** to cancel or reduce common mode emissions. In yet other embodiments, the first channel **501** and the second channel **502** are paired with separate but identical or nearly identical electrical loads (i.e., within five percent (5%) of one another). Doing so may keep the operating conditions and load level the same or nearly the same. As noted, this may enhance the ability of the system **500** to cancel or reduce common mode emissions.
(158) The inventive aspects of the present disclosure may apply to single-channel electrical power systems as well. In this regard, common mode emissions can be canceled or reduced within a single power channel in a similar manner as disclosed above.
(159) By way of example, FIG. **16** provides a schematic view of an electrical power system **600** according to one example embodiment of the present disclosure. As depicted, the power system **600** of FIG. **16** is a single-channel power system. The power system **600** of FIG. **16** is arranged and configured to operate so that common mode emissions can be canceled or reduced within channel **601**. Generally, the power system **600** of FIG. **16** includes a power converter system **610**, an electric machine **650**, and one or more electrical cables **604**, **606** electrically coupling the power converter system **610** and the electric machine **650**. The electric cables **604**, **606** can be shielded cables, for example.
(160) As shown, the power converter system **610** includes first switching elements **611** and second switching elements **612**. The power converter system **610** also includes one or more first gate drivers **621** for driving the first switching elements **611** and one or more second gate drivers **622** for driving the second switching elements **621**. Further, the power converter system **610** includes a DC-side EMI filter **630** and an AC-side EMI filter **632** for suppressing electromagnetic noise transmitted along the channel **601**. The power converter system **610** also includes controller **640**. The controller **640** can be communicatively coupled with various components, such as the gate drivers **621**, **622**. Generally, the controller **640** can be configured and perform operations in the manner illustrated in FIG. **30** and described in the accompanying text.
(161) The electric machine **650** has a rotor (not depicted in FIG. **16**) and a stator **660**. The rotor can be mechanically coupled with a rotating component, such as a rotating component of a gas turbine engine. The rotor is rotatable about an axis of rotation and can include a plurality of magnets. The stator **660** includes a plurality of multiphase windings or coils wound therein, e.g., within slots defined between teeth of the stator **660**. For the embodiment depicted, the stator **660** includes a first multiphase winding **661** and a second multiphase winding **662**. The first multiphase winding **661** and the second multiphase winding **662** are both associated with channel **601**. Each multiphase winding **661**, **662** can include windings or coils for various power phases. Moreover, as depicted schematically in FIG. **16**, the first multiphase winding **661** is electrically coupled with the first switching elements **611** of the power converter system **610** and the second multiphase winding **662** is electrically coupled with the second switching elements **612**.
(162) Notably, the first multiphase winding **661** and the second multiphase winding **662** are electrically opposite in phase with respect to one another. That is, the angle of the AC voltage of the second multiphase winding **662** is electrically out of phase with respect to the angle of the AC voltage of the first multiphase winding **661** by one hundred eighty degrees (180°). For instance, for this embodiment, the first multiphase winding **661** has an AC voltage angle of zero degrees (0°) while the second multiphase winding **662** has an AC voltage angle of one hundred eighty degrees (180°).
(163) Moreover, for this embodiment, the first and second multiphase windings **661**, **662** can be strategically arranged in an interweaved contra-phase winding pair arrangement. Such an

arrangement may facilitate balance of radial forces during operation of the electric machine **650**, among other benefits. As shown, the first multiphase winding **661** is arranged in a first set **665** and a second set **666**. The first set **665** and the second set **666** of the first multiphase winding **661** are positioned opposite one another along the radial direction R. The first set **665** and the second set **666** can be electrically connected to one another in series or parallel.

(164) The second multiphase winding **662** is also arranged in a first set **667** and a second set **668**. The first set **667** and the second set **668** of the second multiphase winding **662** are positioned opposite one another along the radial direction R. The first set **667** and the second set **668** can be electrically connected to one another in series or parallel. As depicted, the first set **665** of the first multiphase winding **661** is positioned between the first set **667** and the second set **668** of the second multiphase winding **662** along the circumferential direction and the second set **666** of the first multiphase winding **661** is positioned between the first set **667** and the second set **668** of the second multiphase winding **662** along the circumferential direction.

(165) Moreover, the stator **660** of the electric machine **650** defines a plurality of sectors or sections. Particularly, for this embodiment, the stator **660** defines four sectors, including a first sector **671**, a second sector **672**, a third sector **673**, and a fourth sector **674**. The sectors **671**, **672**, **673**, **674** are or about of equal size. As shown, the first and second sets **665**, **666** of the first multiphase winding **661** and the first and second sets **667**, **668** of the second multiphase winding **662** are each wound within a respective one of the sectors **671**, **672**, **673**, **674** of the stator **660**. More specifically, the first set **665** of the first multiphase winding **661** is wound within the first sector **671**, the second set **666** of the first multiphase winding **661** is wound within the second sector **672**, the first set **667** of the second multiphase winding **662** is wound within the third sector **673**, and the second set **668** of the second multiphase winding **662** is wound within the fourth sector **674**.

(166) Common mode emissions can be canceled and/or reduced within channel **601** in a similar manner as disclosed above with respect to the first channel **401** or the second channel **402** of the multichannel power system **400** of FIG. **6**. Accordingly, for the sake of brevity, the control scheme in which common mode emissions associated with the power system **600** can be canceled or reduced will not be repeated.

(167) However, to summarize, one or more processors of the controller **640** can receive voltage commands associated with the first multiphase winding **661** and voltage commands associated with the second multiphase winding **662**. The one or more processors of the controller **640** can then generate first control signals based at least in part on the voltage commands associated with the first multiphase winding **661** and second control signals based at least in part on the voltage commands associated with the second multiphase winding **662**. The one or more processors of the controller **640** can control the first switching device **611** based on the first control signals to generate first PWM signals, which renders a first common mode signal. Similarly, the one or more processors of the controller **640** can control the second switching device **612** based on the second control signals to generate second PWM signals, which renders a second common mode signal.

(168) Notably, the rendered second common mode signal can have a same or similar waveform with opposite polarity with respect to the rendered first common mode signal. The common mode signals have the same waveform because the first and second multiphase windings **661**, **662** are electrically opposite in phase with respect to one another and the polarity of the common mode signals are opposite one another due to the one or more processors changing the polarity of the generated second PWM signals, such as by shifting the second carrier signal by one hundred eighty degrees (180°). A number of benefits and advantages associated with such a cancelation technique are possible.

(169) The inventive aspects of the present disclosure may apply to other single-channel electrical power systems as well. By way of example, FIG. **17** provides a schematic view of an electrical power system **700** according to one example embodiment of the present disclosure. As depicted, the power system **700** of FIG. **17** is a single-channel power system. Generally, the power system

700 of FIG. **17** includes a power converter system **710**, an electric machine **750**, and one or more electrical cables **704**, **706** electrically coupling the power converter system **710** and the electric machine **750**. The electric cables **704**, **706** can be shielded cables, for example. For the depicted embodiment of FIG. **17**, the power converter system **710** is configured in the same manner as the power converter system **610** of the power system **600** depicted in FIG. **16**, and thus, the power converter system **710** will not be described in detail.
(170) The electric machine **750** includes a first multiphase winding **761** and a second multiphase windings **762**. The first multiphase winding **761** and the second multiphase winding **762** are arranged and configured to operate electrically opposite in phase with respect to one another. For this embodiment, the first and second multiphase windings **761**, **762** are strategically arranged in a separated contra-phase winding pair arrangement. Such an arrangement may facilitate balance of radial forces during operation of the electric machine **750**, among other benefits. As shown, the first multiphase winding **761** and the second multiphase winding **762** are positioned opposite one another along the radial direction R defined by the electric machine **750**. Particularly, for this embodiment, a stator **760** of the electric machine **750** defines a first half sector **771** and a second half sector **772**. As depicted, the first multiphase winding **761** is wound within the first half sector **771** (e.g., within slots defined by teeth of the stator **760**) and the second the second multiphase winding **762** is wound within the second half sector **772** (e.g., within slots defined by teeth of the stator **760**).
(171) Common mode emissions can be canceled and/or reduced within channel **701** in a similar manner as disclosed above with respect to the first channel **401** or the second channel **402** of the multichannel power system **400** of FIG. **6**. Accordingly, for the sake of brevity, the control scheme in which common mode emissions associated with the power system **700** can be canceled or reduced will not be repeated here.
(172) FIG. **18** provides a schematic view of another single-channel electrical power system **800** according to one example embodiment of the present disclosure. Generally, the power system **800** of FIG. **18** includes a power converter system **810**, an electric machine **850**, and one or more electrical cables **804**, **806** electrically coupling the power converter system **810** and the electric machine **850**. The electric cables **804**, **806** can be shielded cables, for example. For the depicted embodiment of FIG. **18**, the power converter system **810** is configured in the same manner as the power converter system **610** of the power system **600** depicted in FIG. **16**, and thus, the power converter system **810** will not be described in detail.
(173) A first multiphase winding **861** and a second multiphase windings **862** are wound within a stator **860** of the electric machine **850**. The first multiphase winding **861** and the second multiphase winding **862** are arranged and configured to operate electrically opposite in phase with respect to one another. For this embodiment, the first and second multiphase windings **861**, **862** are strategically wound or arranged in a collocated contra-phase arrangement. In some example embodiments, to form the collocated contra-phase arrangement, the first and second multiphase windings **861**, **862** can terminate on opposite ends. In some embodiments, a coil in a slot can be separated into two coils, one for each of the first and second multiphase windings **861**, **862**. In yet other embodiments, the coils in same pole groups can be divided into the two separate first and second multiphase windings **861**, **862**. Such an arrangement may facilitate balance of radial forces during operation of the electric machine **850**, among other benefits.
(174) Common mode emissions can be canceled and/or reduced within channel **801** in a similar manner as disclosed above with respect to the first channel **401** or the second channel **402** of the multichannel power system **400** of FIG. **6**. Accordingly, for the sake of brevity, the control scheme in which common mode emissions associated with the power system **800** can be canceled or reduced will not be repeated here.
(175) FIG. **19** provides a flow diagram for a method (**900**) of operating an electrical power system according to one example embodiment. The method (**900**) can be utilized to operate any of the

example power systems provided herein, for example.

(176) At (**902**), the method (**900**) includes receiving, by one or more processors, one or more voltage commands associated with a first multiphase winding of an electric machine, the first multiphase winding being electrically coupled with first switching elements of a power converter system.

(177) At (**904**), the method (**900**) includes receiving, by the one or more processors, one or more voltage commands associated with a second multiphase winding of the electric machine, the second multiphase winding being electrically coupled with second switching elements of the power converter system. In such implementations, the first multiphase winding and the second multiphase winding are electrically opposite in phase with respect to one another.

(178) At (**906**), the method (**900**) includes controlling, by the one or more processors, the first switching elements to generate one or more first pulse width modulated (PWM) signals so as to render a first common mode signal, the first PWM signals being generated based at least in part on the one or more voltage commands associated with the first multiphase winding. The first PWM signals can include a first PWM signal associated with each phase of the power system, for example.

(179) The first PWM signals can include a first PWM signal associated with each phase of the power system, for example. In controlling the first switching elements to generate the first PWM signals to render the first common mode signal at (**906**), the one or more processors can compare the received voltage commands associated with the first multiphase winding to a first carrier signal. In comparing one of the voltage commands associated with the first multiphase winding to the first carrier signal, in instances in which the voltage command is above the first carrier signal, the first switching elements are controlled so that the generated first PWM signal has a maximum amplitude at that instance. In contrast, in instances in which the voltage command is below the first carrier signal, the first switching elements are controlled so that the generated first PWM signal has a minimum amplitude at that instance. In this manner, the first PWM signals can be generated based at least in part on their associated voltage commands.

(180) At (**908**), the method (**900**) includes controlling, by the one or more processors, the second switching elements to generate one or more second PWM signals so as to render a second common mode signal, the second PWM signals being generated based at least in part on the one or more voltage commands associated with the second multiphase winding, the second common mode signal having opposite polarity with respect to the first common mode signal.

(181) The second PWM signals can include a second PWM signal associated with each phase of the power system. In controlling the second switching elements to generate the second PWM signals to render the second common mode signal at (**908**), the one or more processors can compare the received voltage commands associated with the second multiphase winding to a second carrier signal. Notably, prior to comparing the received voltage commands associated with the second multiphase winding to the second carrier signal, the one or more processors can shift the second carrier signal by one hundred eighty degrees (180°) with respect to the first carrier signal. In this way, the second carrier signal is the inverse of the first carrier signal. In comparing one of the voltage commands associated with the second multiphase winding to the second carrier signal, in instances in which the voltage command is above the second carrier signal, the second switching elements are controlled so that the generated second PWM signal has a maximum amplitude at that instance. In contrast, in instances in which the voltage command is below the second carrier signal, the second switching elements are controlled so that the generated second PWM signal has a minimum amplitude at that instance. In this manner, the second PWM signals can be generated based at least in part on their associated voltage commands.

(182) The pattern or waveform of the first common mode voltage signal is rendered based on the first PWM signals associated with the first multiphase winding. Particularly, for a given instance, the amplitude of the first common mode voltage signal can be an average amplitude of the

generated first PWM signals associated with the first multiphase winding. Similarly, the pattern or waveform of the second common mode voltage signal is rendered based on the second PWM signals associated with the second multiphase winding. Particularly, for a given instance, the amplitude of the second common mode voltage signal can be an average amplitude of the generated PWM signals associated with the second multiphase winding.

(183) In some implementations, the second common mode signal has a same waveform with opposite polarity with respect to the first common mode signal. In this regard, the second common mode signal and the first common mode signal can “mirror” one another. In such implementations, the first and second common mode signals have the same waveform because the first and second multiphase windings are electrically opposite in phase with respect to one another. The polarity of the common mode signals are made opposite one another due to the one or more processors shifting the second carrier signal by one hundred eighty degrees (180°) with respect to the first carrier signal as noted above. As the first and second common mode signals have a same or similar waveform and opposite polarity, common mode emissions can be canceled or reduced.

(184) In some implementations, the first multiphase winding and the second multiphase winding are associated with a same channel of the power system. In other implementations, the first multiphase winding is associated with a first channel of the power system and the second multiphase winding is associated with a second channel of the power system.

(185) In some further implementations, the electric machine defines a radial direction, and wherein the first multiphase winding is arranged in a first set and a second set positioned opposite one another along the radial direction, and wherein the second multiphase winding is arranged in a first set and a second set positioned opposite one another along the radial direction. In such implementations, optionally, the electric machine has a stator defining sectors, and wherein the first and second sets of the first multiphase winding and the first and second sets of the second multiphase winding are each wound within a respective one of the sectors of the stator.

(186) In yet other implementations, the power converter system has third switching elements and fourth switching elements. In such implementations, the electric machine further includes a third multiphase winding electrically coupled with the third switching elements and a fourth multiphase winding electrically coupled with the fourth switching elements, the fourth multiphase winding being electrically opposite in phase with respect to the third multiphase winding. In such implementations, optionally, the electric machine defines a radial direction, and wherein the first multiphase winding is arranged opposite the second multiphase winding along the radial direction and the third multiphase winding is arranged opposite the fourth multiphase winding along the radial direction. In some further implementations, the electric machine has a stator defining sectors, and wherein the first, second, third, and fourth multiphase windings are wound within a respective one of the sectors of the stator.

(187) In some implementations, the electric machine has a stator defining a first half sector and a second half sector, and wherein the first multiphase winding is wound within the first half sector and the second the second multiphase winding is wound within the second half sector.

(188) In some implementations, the first multiphase winding and the second multiphase winding are wound in a collocated contra-phase arrangement.

(189) In some implementations, in response to a detected failure condition associated with the second multiphase winding, the one or more processors are configured to control the second switching elements to synthesize the second common mode signal, e.g., as will be described further below.

(190) In yet other implementations, in response to a detected failure condition associated with the power converter system, the one or more processors are configured to control the second switching elements in one of a single-leg operation mode or a reduced leg operation mode to render the second common mode signal, e.g., as will be described further below.

(191) Notably, the architectures and control schemes disclosed herein can provide a useful degree

of common mode cancellation even when complementary excitation is not possible due to certain conditions, such as load/power unbalance, power-limiting, or failure at the converter or windings of the electric machine.

(192) By way of example and with reference now to FIGS. **20** and **21**, in some instances the power converter system **410** may experience a failure condition. Example failure conditions include, without limitation, open wires or open circuits, line-to-line shorts or circuit shorts, and switching element failures, among other possible failures. For this example, as shown in FIG. **20**, two of the second switching elements **412** electrically coupled with the second multiphase winding **462** of the electric machine **450** have failed (all other multiphase windings **461**, **463**, **464** and their associated switching elements **411**, **413**, **414** are not depicted in FIG. **20**). Particularly, the fourth switching element S4 associated with the B-leg of the power converter system **410** is faulty and the third switching element S3 associated with the C-leg of the power converter system **410** is faulty as well. All switching elements of the A-leg of the power converter system **410** remain operational.

(193) When such a failure condition occurs, the second switching elements **412** can be controlled in a single-leg operation mode to cancel, at least in part, the common mode voltage associated with the first switching elements **411** and first multiphase winding **461** (FIG. **8**). More specifically, the second switching elements **412** can be controlled to produce a common mode voltage that partially cancels the common mode voltage produced by operation of the first switching elements **411** (FIG. **8**). For instance, as depicted in the graph of FIG. **21**A, the first common mode signal Vcm1 associated with the first multiphase winding **461** and first switching elements **411** is depicted. The first common mode signal Vcm1 can be generated based at least in part on first PWM signals as described above.

(194) With reference also to FIG. **9** in addition to FIGS. **21**A through **21**C, upon executing the second modulator **442** of the pulse width modulator **448**, the one or more processors of the controller **440** can generate a second PWM signal associated with the A phase of the power system **400** (FIG. **6**), denoted as PWM-Va2 in the graph of FIG. **21**B, e.g., based at least in part on voltage command Va2. As the B-leg and C-leg of the power converter system **410** associated with the second multiphase winding **462** and second switching elements **412** are shutdown or non-operational due to failure of the fourth switching element S4 of the B-leg and the third switching element S3 of the C-leg, no second PWM signals are generated for the B and C phases.

(195) Accordingly, the second common mode signal Vcm2 depicted in the graph of FIG. **21**C, which is rendered based at least in part on the generated second PWM signals, has the same waveform as the second PWM signal PWM-Va2. As will be appreciated by comparing the first common mode signal Vcm1 depicted in the graph of FIG. **21**A with the second common mode signal Vcm2 depicted in the graph of FIG. **21**C, the second common mode signal Vcm2 has opposite polarity but not the exact waveform with respect to the first common mode signal Vcm1. The waveforms of the first and second common mode signals Vcm1, Vcm2 are not exact in waveform because the failure condition has caused the B and C windings of the second multiphase winding **462** and their associated circuitry to be the same voltage as the A phase winding and circuitry. Accordingly, PWM signals associated with the B and C phases of the second multiphase winding **462** and second switching elements **412** are not generated or are not impactful in rendering the second common mode voltage signal Vcm2. The polarity of the second common mode signal Vcm2 is shifted with respect to the first common mode voltage signal Vcm1, and hence, the first common mode signal Vcm1 and the second common mode signal Vcm2 have opposite polarity. Advantageously, the common mode voltage associated with the first switching elements **411** and first multiphase winding **461** may still be canceled at least in part by the common mode voltage associated with the second switching elements **412** and second multiphase winding **462**. It will be appreciated that these inventive aspects may be apply to other embodiments disclosed herein, such as the embodiments depicted in FIGS. **14** and **16-18**.

(196) As another example and with reference now to FIGS. **22** and **23**, in some instances the power

system **400** may experience a failure condition where at least one leg of the power converter system **410** has failed but multiple legs remain operational. For instance, as shown in FIG. **22**, one of the second switching elements **412** electrically coupled with the second multiphase winding **462** of the electric machine **450** has failed (all other multiphase windings **461**, **463**, **464** and their associated switching elements **411**, **413**, **414** are not depicted in FIG. **22**). Specifically, the first switching element S1 associated with the C-leg of the power converter system **410** is faulty. All switching elements of the A-leg and the B-leg of the power converter system **410** associated with the second multiphase winding **462** remain operational.

(197) When such a failure condition occurs, the second switching elements **412** can be controlled in a multi-leg operation mode to cancel, at least in part, the common mode voltage associated with the first switching elements **411** and first multiphase winding **461** (FIG. **8**). More specifically, the second switching elements **412** can be controlled to produce a common mode voltage that partially cancels the common mode voltage produced by operation of the first switching elements **411** (FIG. **8**). For instance, as depicted in the graph of FIG. **23**A, the first common mode signal Vcm1 associated with the first multiphase winding **461** and first switching elements **411** is depicted. The first common mode signal Vcm1 can be generated based at least in part on first PWM signals as described previously.

(198) With reference also to FIG. **9** in addition to FIGS. **23**A through **23**D, upon executing the second modulator **442** of the pulse width modulator **448**, the one or more processors of the controller **440** can generate a second PWM signal associated with an A phase of the power system **400** (FIG. **6**), denoted as PWM-Va2 in the graph of FIG. **23**B, e.g., based at least in part on voltage command Va2. Similarly, upon executing the second modulator **442** of the pulse width modulator **448**, the one or more processors of the controller **440** can generate a second PWM signal associated with a B phase of the power system **400**, denoted as PWM-Vb2 in the graph of FIG. **23**C, e.g., based at least in part on voltage command Vb2. As the C-leg of the power converter system **410** associated with the second multiphase winding **462** and second switching elements **412** is shutdown or non-operational due to failure of the first switching element S1 of the B-leg, no second PWM signal is generated for the C phase.

(199) Accordingly, the second common mode signal Vcm2 depicted in the graph of FIG. **23**D is rendered based at least in part on second PWM signal PWM-Va2 and second PWM signal PWM-Vb2. For a given point in time, the amplitude of the second common mode voltage signal Vcm2 is an average amplitude of the second PWM signal PWM-Va2 and the second PWM signal PWM-Vb2 associated with the second multiphase winding **462**.

(200) As will be appreciated by comparing the first common mode voltage signal Vcm1 depicted in the graph of FIG. **23**A with the second common mode signal Vcm2 depicted in the graph of FIG. **23**D, the second common mode signal Vcm2 has opposite polarity but with a similar but not exact waveform with respect to the first common mode signal Vcm1. The waveforms of the first and second common mode signals Vcm1, Vcm2 are similar in waveform because the first multiphase winding **461** and the second multiphase winding **462** are arranged and configured to operate electrically opposite in phase with respect to one another but not exact in waveform because of the failure condition associated with the C phase. The first common mode signal Vcm1 is a four-level waveform and the second common mode signal Vcm2 is a three-level waveform in this example. The polarity of the second common mode signal Vcm2 is shifted with respect to the first common mode voltage signal Vcm1, and hence, the first common mode signal Vcm1 and the second common mode signal Vcm2 have opposite polarity. Advantageously, the common mode voltage associated with the first switching elements **411** and first multiphase winding **461** may still be canceled at least in part by the common mode voltage associated with the second switching elements **412** and second multiphase winding **462** due to the opposite polarity and similarity in waveform of the rendered first and second common mode signals Vcm1, Vcm2. In some instances, significant cancelation is possible. It will be appreciated that these noted inventive aspects may be

apply to other embodiments disclosed herein, such as the embodiments depicted in FIGS. **14** and **16-18**.

(201) In some instances, the power system may experience a failure condition where one multiphase winding fails but yet its complementary multiphase winding is still operable. By way of example and with reference to FIG. **14**, the second multiphase winding **562** associated with the second channel **502** may become inoperable while the first multiphase winding **561** associated with the first channel **501** may be operable, or vice versa. In such instances, the switching elements associated with the inoperable multiphase winding can be controlled to synthesize a common mode signal that can cancel, at least in part, the common mode signal associated with the operable multiphase winding. In this regard, the common mode voltage associated with the operable multiphase winding can be canceled at least in part by the common mode voltage associated with the inoperable multiphase winding.

(202) Particularly, suppose the power system **500** is functioning normally during operation. Then, the second multiphase winding **562** experiences a failure condition. In response to the detected failure condition associated with the second multiphase winding **562**, the one or more processors of the controller **540** can control the second switching elements **512** to synthesize a second common mode signal having opposite polarity with respect to a first common mode signal associated with the first switching elements **511** and first multiphase winding **561**. The failure condition associated with the second multiphase winding **562** can be detected in any suitable manner, such as by temperature sensors associated with the multiphase windings **561**, **562** and/or power converter system **510**, electric current and/or voltage sensors on the AC and/or DC side of the power converter system **510**, speed sensors, baseline models, a combination of the foregoing, etc.

(203) In controlling the second switching elements **512** to synthesize the second common mode signal, the one or more processors of the controller **540** are configured to control the second switching elements **512** to generate second PWM signals. Each of the second PWM signals can be associated with a respective phase of the power system **500**. With reference also to FIG. **15** in addition to FIGS. **14** and **23**, upon executing the second modulator **542** of the pulse width modulator **548**, the one or more processors of the controller **540** can generate the second PWM signals. Specifically, the one or more processors of the controller **540** can generate a second PWM signal associated with an A-phase of the power system **500**, denoted in the graph of FIG. **24**B as PWM-Va2, a second PWM signal associated with a B-phase of the power system **500**, denoted in the graph of FIG. **24**C as PWM-Vb2, and a second PWM signal associated with a C-phase of the power system **500**, denoted in the graph of FIG. **24**D as PWM-Vc2.

(204) In some example embodiments, the duty cycles of the second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 can be selected and the second switching elements **512** can be controlled so that each one of the second PWM signals has a duty cycle of fifty percent (50%), e.g., as shown in the graphs of FIGS. **24**B, **24**C, **24**D. At a duty cycle of fifty percent (50%), each second PWM signal is on or “high” for fifty percent of its duty cycle and is off or “low” for the remaining fifty percent of its cycle. Accordingly, the second PWM signals are generated as square waves.

(205) Due to the selected duty cycles of the second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2, the one or more processors of the controller **540** can control the second switching elements **512** to generate the second PWM signals so that each terminal positioned between the second switching elements **512** and the second multiphase winding **562** has a voltage that averages about zero volts. An A-phase terminal TA associated with the A-phase of the power system **400**, a B-phase terminal TB associated with the B-phase of the power system **400**, and a C-phase terminal TC associated with the C-phase of the power system **400** are depicted in FIG. **14**. By minimizing the voltage associated with the now-faulty second multiphase winding **562**, e.g., to zero volts on average, the second switching elements **512** can be controlled to rapidly modulate to generate the second PWM signals to render the second common mode signal Vcm2, but notably, can do so without converting electrical power. This may, for example, cause the second multiphase winding

562 to cease producing torque, which may be beneficial due to the failure condition associated with the second multiphase winding **562**.
(206) In addition, the one or more processors can control the second switching elements **512** to generate the second PWM signals so that a polarity of each of the second PWM signals is opposite with respect to a waveform of the first common mode signal Vcm1, which is depicted in the graph of FIG. **24**A. In this way, the rendered second common mode signal Vcm2, which is depicted in the graph of FIG. **24**E, has opposite polarity with respect to the first common mode signal Vcm1. This facilitates cancelation of the common mode voltage.
(207) Moreover, under a first synthesis control scheme, in controlling the second switching elements **512** to synthesize the second common mode signal Vcm2, the one or more processors of the controller **540** are configured to control the second switching elements **512** to generate the second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 so that the second PWM signal PWM-Va2 associated with the first phase of the power system **400** (e.g., the A phase), the second PWM signal PWM-Vb2 associated with the second phase of the power system **400** (e.g., the B phase), and the second PWM signal PWM-Vc2 associated with the third phase of the power system **400** (e.g., the C phase), are offset from one another in phase.
(208) For instance, for the second PWM signals depicted in the close-up views of the graphs of FIGS. **24**B, **24**C, **24**D, the second PWM signals associated with the first, second, and third phases of the power system **400** are generated so that the second PWM signals are offset in phase from one another by about one hundred twenty degrees (120°). That is, the second PWM signal PWM-Va2 is offset from the second PWM signal PWM-Vb2 and the second PWM signal PWM-Vc2 by about one hundred twenty degrees (120°). Similarly, the second PWM signal PWM-Vb2 is offset from the second PWM signal PWM-Va2 and the second PWM signal PWM-Vc2 by about one hundred twenty degrees (120°). Further, as can be deduced, the second PWM signal PWM-Vc2 is offset from the second PWM signal PWM-Va2 and the second PWM signal PWM-Vb2 by about one hundred twenty degrees (120°). Offsetting the second PWM signals, e.g., by about one hundred twenty degrees (120°), produces a resultant or rendered second common mode signal Vcm2 (depicted in the graph of FIG. **24**E) that has a similar waveform to the first common mode signal Vcm1 (depicted in the graph of FIG. **24**A). For instance, offsetting the second PWM signals, e.g., by about one hundred twenty degrees (120°), produces a resultant or rendered second common mode signal Vcm2 that is a multilevel signal (four levels in this embodiment) that matches the number of levels of the first common mode signal Vcm1.
(209) Further, the plurality of second PWM signals are generated so as to each have a same or about an equivalent voltage amplitude, e.g., as shown in the graphs of FIGS. **24**B, **24**C, **24**D. In this way, the second multiphase winding **562** of the electric machine **550** may not experience significant differential voltage, and consequently, differential currents are negligible. Thus, for this embodiment, the plurality of second PWM signals are generated so as to each have a same duty cycle, about an equivalent voltage amplitude, and so as to be out-of-phase or offset in phase with respect to one another.
(210) Under a second synthesis control scheme, in controlling the second switching elements **512** to synthesize the second common mode signal Vcm2, the one or more processors of the controller **540** are configured to control the second switching elements **512** to generate the second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 so that the second PWM signal PWM-Va2 associated with the first phase of the power system **400** (e.g., the A phase), the second PWM signal PWM-Vb2 associated with the second phase of the power system **400** (e.g., the B phase), and the second PWM signal PWM-Vc2 associated with the third phase of the power system **400** (e.g., the C phase), are in phase with one another.
(211) For instance, for the second PWM signals depicted in the close-up views of the graphs of FIGS. **25**B, **25**C, **25**D, the second PWM signals associated with the first, second, and third phases of the power system **400** are generated so that the second PWM signals are in phase with one

another. That is, the second PWM signal PWM-Va2, the second PWM signal PWM-Vb2, and the second PWM signal PWM-Vc2 are in phase with one another. Moreover, the generated second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 each have the same duty cycle, which is a fifty percent (50%) duty cycle in this example embodiment. In this way, the plurality of second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 are generated so as to each have a same duty cycle and so as to be in phase with one another. Accordingly, the second PWM signals depicted in the graphs of FIGS. **25**B, **25**C, **25**D produce a resultant or rendered second common mode signal Vcm2 (depicted in the graph of FIG. **25**E) that has a same waveform as each of the second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 depicted in the graphs of FIGS. **25**B, **25**C, **25**D. Further, the plurality of second PWM signals are generated so as to each have a same or about an equivalent voltage amplitude. In this way, the second multiphase winding **562** of the electric machine **550** may not experience significant differential voltage, and consequently, differential currents are negligible. Thus, for this embodiment, the plurality of second PWM signals are generated so as to each have a same duty cycle, about an equivalent voltage amplitude, and so as to be in phase with one another.

(212) Although the second common mode signal Vcm2 depicted in the graph of FIG. **25**E is somewhat different in waveform with respect to the first common mode signal Vcm1 depicted in the graph of FIG. **25**A, the second common mode signal Vcm2 cancels the first common mode signal Vcm1 at least in part, thereby providing a degree of common mode cancelation.

(213) FIGS. **26**A, **26**B, and **26**C each show a graph depicting voltage signals as a function of time. Particularly, the graph of FIG. **26**A shows the first common mode signal Vcm1 rendered as per normal operation in which the first multiphase winding **561** is operable and depicts the second common mode signal Vcm2 rendered without implementation of the first or second synthesis control scheme when the second multiphase winding **562** has experienced a failure condition. As depicted, the sum of the first common mode signal Vcm1 and the second common mode signal Vcm2 essentially matches the waveform of the first common mode signal Vcm1. Consequently, there is effectively no cancelation of common mode emissions when synthesis control techniques are not implemented.

(214) The graph of FIG. **26**B shows the first common mode signal Vcm1 rendered as per normal operation in which the first multiphase winding **561** is operable and depicts the second common mode signal Vcm2 rendered in accordance with the first synthesis control scheme in response to the failure condition associated with the second multiphase winding **562**. As depicted, the generated second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 are time-phase shifted thus rendering a four-level second common mode signal Vcm2. As illustrated, the first common mode signal Vcm1 is largely “mirrored” by the second common mode signal Vcm2, which provides significant cancelation of common mode emissions. The sum of the first common mode signal Vcm1 and the second common mode signal Vcm2 is largely a flat, constant line but does have more noise than the ideal common mode cancelation scenario depicted in FIG. **13**.

(215) The graph of FIG. **26**C shows the first common mode signal Vcm1 rendered as per normal operation in which the first multiphase winding **561** is operable and depicts the second common mode signal Vcm2 rendered in accordance with the second synthesis control scheme in response to the failure condition associated with the second multiphase winding **562**. As depicted, the generated second PWM signals PWM-Va2, PWM-Vb2, PWM-Vc2 are not time-phase shifted. That is, the second PWM signals are all in phase with one another. As a result, the rendered second common mode signal Vcm2 is only a two-level signal. The sum of the first common mode signal Vcm1 and the second common mode signal Vcm2 is largely a flat, constant line but does have more noise than the ideal case shown in FIG. **13** and more noise than the summation of the first common mode signal Vcm1 and the second common mode signal Vcm2 under the first synthesis control scheme depicted in FIG. **26**B. However, an effective degree of common mode cancelation may be achieved as shown in FIG. **26**C under the second synthesis control scheme.

(216) While implementing the first synthesis control scheme may provide more effective common mode cancelation than implementation of the second synthesis control scheme when a multiphase winding has suffered a failure condition, implementation of the second synthesis control scheme can prevent or reduce high frequency current ripples within the power system **500**. Accordingly, in some instances, the inventors of the present disclosure have discovered that it may be advantageous to switch between the first and second synthesis control schemes.
(217) For instance, in response to a detected failure condition associated with the second multiphase winding **562**, the second switching elements **512** can be controlled to synthesize the second common mode signal Vcm2 using the first synthesis control scheme as a default scheme. However, in response to determining that a high frequency ripple current within the power system **400** has reached a threshold, the one or more processors of the controller **540** can switch from implementing the first control scheme to the second control scheme at least until a target condition is met. The target condition can be, without limitation, a time period, a predetermined safety threshold at which the high frequency ripple current is deemed acceptable, etc. As one example, in response to determining that a high frequency ripple current within the power system **400** has reached a threshold, the one or more processors of the controller **540** can switch from implementing the first control scheme to the second control scheme at least until a predetermined time period (i.e., the target condition) has elapsed. As another example, in response to determining that a high frequency ripple current within the power system **400** has reached a threshold, the one or more processors of the controller **540** can switch from implementing the first control scheme to the second control scheme at least until the high frequency ripple current has reached a predetermined safety threshold (i.e., the target condition). When the target condition is met, the one or more processors of the controller **540** can resume implementing the first synthesis control scheme.
(218) Accordingly, switching between the first and second synthesis control schemes can allow for effective cancelation of common mode emissions, and if necessary, prevention or reduction of high frequency ripple current despite the second multiphase winding **562** failure condition.
(219) It will be appreciated that the inventive aspects relating to control schemes associated with cancelation of common mode emissions where one multiphase winding fails but yet its complementary multiphase winding is still operable may be apply to other embodiments disclosed herein, such as the embodiments depicted in FIGS. **6** and **16-18**. Particularly, in the embodiment of FIG. **6**, the above-noted inventive aspects can be applied within a single channel, e.g., where first multiphase winding **461** is operable and second multiphase winding **462** is inoperable or vice versa.
(220) FIG. **27** provides a flow diagram for a method (**920**) of operating an electrical power system according to one example embodiment. The method (**920**) can be utilized to operate any of the example power systems provided herein, for example.
(221) At (**922**), the method (**920**) includes receiving, by one or more processors, one or more voltage commands associated with a first multiphase winding of an electric machine, the first multiphase winding being electrically coupled with first switching elements of a power converter system, the electric machine having a second multiphase winding being electrically coupled with second switching elements of the power converter system, the first multiphase winding and the second multiphase winding being electrically opposite in phase with respect to one another.
(222) At (**924**), the method (**920**) includes controlling, by the one or more processors, the first switching elements to generate one or more first pulse width modulated (PWM) signals so as to render a first common mode signal, the first PWM signals being generated based at least in part on the one or more voltage commands associated with the first multiphase winding.
(223) At (**926**), the method (**920**) includes in response to a detected failure condition associated with the second multiphase winding, controlling, by the one or more processors, the second switching elements to synthesize a second common mode signal having opposite polarity with respect to the first common mode signal.

(224) In some implementations, controlling the second switching elements to synthesize the second common mode signal includes controlling, by the one or more processors, the second switching elements to generate second PWM signals. For instance, a second PWM signal can be generated for each phase of the power system. The second PWM signals can be generated so that a polarity of each of the second PWM signals is opposite with respect to a waveform of the first common mode signal.
(225) In some further implementations, the second switching elements are controlled by the one or more processors to generate the second PWM signals so that each terminal positioned between the second switching elements and the second multiphase winding has a voltage that averages about zero volts. In yet other implementations, the second switching elements are controlled to generate the second PWM signals so that each one of the second PWM signals has a duty cycle of fifty percent or about 50%.
(226) In some implementations, the one or more second PWM signals include a second PWM signal associated with a first phase of the power system, a second PWM signal associated with a second phase of the power system, and a second PWM signal associated with a third phase of the power system, and wherein, in controlling the second switching elements to generate the second PWM signals, the one or more processors implement a first synthesis control scheme. Under the first synthesis control scheme, the one or more processors shift at least two of: the second PWM signal associated with the first phase of the power system, the second PWM signal associated with the second phase of the power system, and the second PWM signal associated with the third phase of the power system so that the second PWM signals associated with the first, second, and third phases of the power system are offset from one another in phase, e.g., as shown in FIGS. **24**A through **24**E and FIG. **26**B. In such implementations, optionally, the at least two of the second PWM signals associated with the first, second, and third phases of the power system are shifted so that the second PWM signals associated with the first, second, and third phases of the power system are offset in phase from one another by about one hundred twenty degrees.
(227) In some implementations, the one or more second PWM signals include a plurality of second PWM signals each being associated with different phases of the power system, and wherein in controlling the second switching elements to generate the second PWM signals, the one or more processors implement a second synthesis control scheme. Under the second synthesis control scheme, the one or more processors generate the plurality of second PWM signals so as to be in phase with one another, e.g., as shown in FIGS. **25**A through **25**E and FIG. **26**C.
(228) In some implementations, the one or more second PWM signals include a plurality of second PWM signals each being associated with different phases of the power system, and wherein the plurality of second PWM signals are generated so as to each have a same duty cycle, about an equivalent voltage amplitude, and so as to be in phase with one another.
(229) In some further implementations, in controlling the second switching elements to synthesize the second common mode signal in response to the detected failure condition associated with the second multiphase winding, the one or more processors are configured to, in response to determining that a high frequency ripple current within the power system has reached a threshold, switch from implementing a first synthesis control scheme to a second synthesis control scheme at least until a target condition is met. In such implementations, under the first synthesis control scheme, the second switching elements are controlled by the one or more processors to generate the second PWM signals so as to be offset in phase from one another. Under the second synthesis control scheme, the second switching elements are controlled by the one or more processors to generate the second PWM signals so as to be in phase with one another.
(230) Although the inventive aspects have been described and illustrated herein in the context of electrical power systems having radial flux rotating electric machines, the inventive aspects also apply to electrical power systems having axial flux rotating electric machines. Examples are provided below.

(231) FIG. **28** provides a schematic cross-sectional view of one example axial flux rotating electric machine **450** according to an example embodiment of the present disclosure. The electric machine **450** of FIG. **28** may be incorporated into the power system **400** of FIG. **6**, for example. As depicted in FIG. **28**, the electric machine **450** includes a rotor **452** and a stator, which includes a first stator **460**A and a second stator **460**B. The rotor **452** is mechanically coupled with a shaft, which is low speed shaft **246** in this example embodiment. The rotor **452** is rotatable about the axis of rotation AX in unison with the low speed shaft **246**. The first stator **460**A is positioned on a first side of the rotor **452** and the second stator **460**B is positioned on a second side of the rotor **452** opposite the first side.
(232) The first stator **460**A has associated first and third multiphase windings **461**, **463** and the second stator **460**B has associated second and fourth multiphase windings **462**, **464**. The first and second multiphase windings **461**, **462** are associated with a first channel (e.g., first channel **401** depicted in FIG. **6**) and the third and fourth multiphase windings **463**, **464** are associated with a second channel (e.g., second channel **402** depicted in FIG. **6**). The first multiphase winding **461** and the second multiphase winding **462** are electrically opposite in phase with respect to one another. Similarly, the third multiphase winding **463** and the fourth multiphase winding **464** are electrically opposite in phase with respect to one another. Notably, the first multiphase winding **461** of the first stator **460**A is positioned opposite the second multiphase winding **462** of the second stator **460**B. In addition, the third multiphase winding **463** of the first stator **460**A is positioned opposite the fourth multiphase winding **464** of the second stator **460**B. Advantageously, the arrangement of the multiphase windings **461**, **462**, **463**, **464** of the electric machine **450** in FIG. **28** allows for the net axial force to be equal to or about 0 during operation, even when only one of the channels is operating.
(233) FIG. **29** provides a schematic cross-sectional view of another example axial flux rotating electric machine **550** according to an example embodiment of the present disclosure. The electric machine **550** of FIG. **29** may be incorporated into the power system **500** of FIG. **14**, for example. As depicted in FIG. **29**, the electric machine **550** includes a rotor **552** and a stator, which includes a first stator **560**A and a second stator **560**B. The rotor **552** is mechanically coupled with a shaft, which is low speed shaft **246** in this example embodiment. The rotor **552** is rotatable about the axis of rotation AX in unison with the low speed shaft **246**. The first stator **560**A is positioned on a first side of the rotor **552** and the second stator **560**B is positioned on a second side of the rotor **552** opposite the first side.
(234) For this embodiment, the electric machine **550** includes a first multiphase winding **561** associated with a first channel (e.g., first channel **501** in FIG. **14**) and a second multiphase winding **562** associated with a second channel (e.g., second channel **502** in FIG. **14**). The first multiphase winding **561** and the second multiphase winding **562** are electrically opposite in phase with respect to one another. The first multiphase winding **561** is arranged in a first set **565** and a second set **566**. The first set **565** and the second set **566** can be electrically connected to one another in series or parallel. Like the first multiphase winding **561**, the second multiphase winding **562** is arranged in a first set **567** and a second set **568**. The first set **567** and the second set **568** can be electrically connected to one another in series or parallel.
(235) As depicted in FIG. **29**, the first stator **560**A includes the first set **565** of the first multiphase winding **561** and the first set **567** of the second multiphase winding **562**. The second stator **560**B includes the second set **566** of the first multiphase winding **561** and the second set **568** of the second multiphase winding **562**. The first set **565** and the second set **566** of the first multiphase winding **561** are spaced from one another along the axial direction A and are positioned opposite one another along the radial direction R. The first set **567** and the second set **568** of the second multiphase winding **562** are spaced from one another along the axial direction A and are positioned opposite one another along the radial direction R.
(236) Advantageously, such a balanced arrangement of the first and second multiphase windings

561, **562** allows for the net axial force associated with the electric machine **550** to be equal to or about 0 during operation. Moreover, when both channels are in operation, the first set **565** and the second set **566** of the first multiphase winding **561** balance out the radial forces therebetween and the first set **567** and the second set **568** of the second multiphase winding **562** balance out the radial forces therebetween. In addition, the balanced arrangement allows for one channel to continue operating at its full power even in the event of non-use or failure of the other channel.

(237) FIG. **30** provides an example computing system **950** according to example embodiments of the present disclosure. The computing devices or elements described herein, such as controller **440**, controller **540**, controller **640**, controller **740**, and controller **840**, may include various components and perform various functions of the computing system **950** described below, for example.

(238) As shown in FIG. **30**, the computing system **950** can include one or more computing device(s) **960**. The computing device(s) **960** can include one or more processor(s) **960**A and one or more memory device(s) **960**B. The one or more processor(s) **960**A can include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, logic device, and/or other suitable processing device. The one or more memory device(s) **960**B can include one or more computer-executable or computer-readable media, including, but not limited to, non-transitory computer-readable media, RAM, ROM, hard drives, flash drives, and/or other memory devices.

(239) The one or more memory device(s) **960**B can store information accessible by the one or more processor(s) **960**A, including computer-readable instructions **960**C that can be executed by the one or more processor(s) **960**A. The instructions **960**C can be any set of instructions that when executed by the one or more processor(s) **960**A, cause the one or more processor(s) **960**A to perform operations. In some embodiments, the instructions **960**C can be executed by the one or more processor(s) **960**A to cause the one or more processor(s) **960**A to perform operations, such as any of the operations and functions for which the computing system **950** and/or the computing device(s) **960** are configured, such as controlling operation of electrical power systems. The instructions **960**C can be software written in any suitable programming language or can be implemented in hardware. Additionally, and/or alternatively, the instructions **960**C can be executed in logically and/or virtually separate threads on processor(s) **960**A. The memory device(s) **960**B can further store data **960**D that can be accessed by the processor(s) **960**A. For example, the data **960**D can include models, lookup tables, databases, etc.

(240) The computing device(s) **960** can also include a network interface **960**E used to communicate, for example, with the other components of system **950** (e.g., via a communication network). The network interface **960**E can include any suitable components for interfacing with one or more network(s), including for example, transmitters, receivers, ports, controllers, antennas, and/or other suitable components. One or more devices can be configured to receive one or more commands from the computing device(s) **960** or provide one or more commands to the computing device(s) **960**.

(241) The technology discussed herein makes reference to computer-based systems and actions taken by and information sent to and from computer-based systems. One of ordinary skill in the art will recognize that the inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. For instance, processes discussed herein can be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications can be implemented on a single system or distributed across multiple systems. Distributed components can operate sequentially or in parallel.

(242) Although specific features of various embodiments may be shown in some drawings and not in others, this is for convenience only. In accordance with the principles of the present disclosure, any feature of a drawing may be referenced and/or claimed in combination with any feature of any other drawing.

(243) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.
(244) Further aspects of the invention are provided by the subject matter of the following clauses:
(245) 1. A power system, comprising: a power converter system having first switching elements and second switching elements; an electric machine, comprising: a first multiphase winding electrically coupled with the first switching elements; and a second multiphase winding electrically coupled with the second switching elements, the first multiphase winding and the second multiphase winding being electrically opposite in phase with respect to one another; and one or more processors configured to: control the first switching elements to generate one or more first pulse width modulated (PWM) signals so as to render a first common mode signal, the first PWM signals being generated based at least in part on one or more voltage commands associated with the first multiphase winding; and control the second switching elements to generate one or more second PWM signals so as to render a second common mode signal, the second PWM signals being generated based at least in part on one or more voltage commands associated with the second multiphase winding, the second common mode signal having opposite polarity with respect to the first common mode signal.
(246) 2. The power system of any preceding clause, wherein the second common mode signal has a same waveform as the first common mode signal.
(247) 3. The power system of any preceding clause, wherein in generating the one or more second PWM signals, the one or more processors are configured to: shift a polarity of each one of the one or more second PWM signals.
(248) 4. The power system of any preceding clause, wherein the first multiphase winding and the second multiphase winding are associated with a same channel of the power system.
(249) 5. The power system of any preceding clause, wherein the first multiphase winding is associated with a first channel of the power system and the second multiphase winding is associated with a second channel of the power system.
(250) 6. The power system of any preceding clause, wherein the electric machine defines a radial direction, and wherein the first multiphase winding is arranged in a first set and a second set positioned opposite one another along the radial direction, and wherein the second multiphase winding is arranged in a first set and a second set positioned opposite one another along the radial direction.
(251) 7. The power system of any preceding clause, wherein the electric machine has a stator defining sectors, and wherein the first and second sets of the first multiphase winding and the first and second sets of the second multiphase winding are each wound within a respective one of the sectors of the stator.
(252) 8. The power system of any preceding clause, wherein the power converter system has third switching elements and fourth switching elements, and wherein the electric machine further comprises: a third multiphase winding electrically coupled with the third switching elements; and a fourth multiphase winding electrically coupled with the fourth switching elements, the fourth multiphase winding being electrically opposite in phase with respect to the third multiphase winding.
(253) 9. The power system of any preceding clause, wherein the electric machine defines a radial direction, and wherein the first multiphase winding is arranged opposite the second multiphase winding along the radial direction and the third multiphase winding is arranged opposite the fourth multiphase winding along the radial direction.

(254) 10. The power system of any preceding clause, wherein the electric machine has a stator defining sectors, and wherein the first, second, third, and fourth multiphase windings are wound within a respective one of the sectors of the stator.
(255) 11. The power system of any preceding clause, wherein the electric machine has a stator defining a first half sector and a second half sector, and wherein the first multiphase winding is wound within the first half sector and the second multiphase winding is wound within the second half sector.
(256) 12. The power system of any preceding clause, wherein the first multiphase winding and the second multiphase winding are wound in a collocated contra-phase arrangement.
(257) 13. The power system of any preceding clause, wherein the electric machine has a rotor and a stator, and wherein the rotor is mechanically coupled with a rotating component of a gas turbine engine.
(258) 14. The power system of any preceding clause, wherein, in response to a detected failure condition associated with the second multiphase winding, the one or more processors are configured to: control the second switching elements to synthesize the second common mode signal.
(259) 15. The power system of any preceding clause, wherein, in response to a detected failure condition associated with the power converter system, the one or more processors are configured to: control the second switching elements in one of a single-leg operation mode or a reduced leg operation mode to render the second common mode signal.
(260) 16. A method, comprising: receiving, by one or more processors, one or more voltage commands associated with a first multiphase winding of an electric machine, the first multiphase winding being electrically coupled with first switching elements of a power converter system; receiving, by the one or more processors, one or more voltage commands associated with a second multiphase winding of the electric machine, the second multiphase winding being electrically coupled with second switching elements of the power converter system, the first multiphase winding and the second multiphase winding being electrically opposite in phase with respect to one another; controlling, by the one or more processors, the first switching elements to generate one or more first pulse width modulated (PWM) signals so as to render a first common mode signal, the first PWM signals being generated based at least in part on the one or more voltage commands associated with the first multiphase winding; and controlling, by the one or more processors, the second switching elements to generate one or more second PWM signals so as to render a second common mode signal, the second PWM signals being generated based at least in part on the one or more voltage commands associated with the second multiphase winding, the second common mode signal having opposite polarity with respect to the first common mode signal.
(261) 17. A power system, comprising: a power converter system having first switching elements and second switching elements; an electric machine, comprising: a first multiphase winding electrically coupled with the first switching elements; and a second multiphase winding electrically coupled with the second switching elements, the first multiphase winding and the second multiphase winding being electrically opposite in phase with respect to one another; and one or more processors configured to: control the first switching elements to generate one or more first pulse width modulated (PWM) signals so as to render a first common mode signal, the first PWM signals being generated based at least in part on one or more voltage commands associated with the first multiphase winding; and in response to a detected failure condition associated with the second multiphase winding, control the second switching elements to synthesize a second common mode signal having opposite polarity with respect to the first common mode signal.
(262) 18. The power system of any preceding clause, wherein in controlling the second switching elements to synthesize the second common mode signal, the one or more processors are configured to control the second switching elements to generate second PWM signals.
(263) 19. The power system of any preceding clause, wherein the one or more processors control

the second switching elements to generate the second PWM signals so that a polarity of each of the second PWM signals is opposite with respect to a waveform of the first common mode signal.
(264) 20. The power system of any preceding clause, wherein the one or more processors control the second switching elements to generate the second PWM signals so that each terminal positioned between the second switching elements and the second multiphase winding has a voltage that averages about zero volts.
(265) 21. The power system of any preceding clause, wherein the one or more processors control the second switching elements to generate the second PWM signals so that each one of the second PWM signals has a duty cycle of fifty percent.
(266) 22. The power system of any preceding clause, wherein the one or more second PWM signals include a second PWM signal associated with a first phase of the power system, a second PWM signal associated with a second phase of the power system, and a second PWM signal associated with a third phase of the power system, and wherein, in controlling the second switching elements to generate the second PWM signals, the one or more processors are further configured to: shift at least two of: the second PWM signal associated with the first phase of the power system, the second PWM signal associated with the second phase of the power system, and the second PWM signal associated with the third phase of the power system so that the second PWM signals associated with the first, second, and third phases of the power system are offset from one another in phase.
(267) 23. The power system of any preceding clause, wherein the at least two of the second PWM signals associated with the first, second, and third phases of the power system are shifted so that the second PWM signals associated with the first, second, and third phases of the power system are offset in phase from one another by about one hundred twenty degrees.
(268) 24. The power system of any preceding clause, wherein the one or more second PWM signals include a plurality of second PWM signals each being associated with different phases of the power system, and wherein the plurality of second PWM signals are generated so as to be in phase with one another.
(269) 25. The power system of any preceding clause, wherein the one or more second PWM signals include a plurality of second PWM signals each being associated with different phases of the power system, and wherein the plurality of second PWM signals are generated so as to each have a same duty cycle, about an equivalent voltage amplitude, and so as to be in phase with one another.
(270) 26. The power system of any preceding clause, wherein in controlling the second switching elements to synthesize the second common mode signal in response to the detected failure condition associated with the second multiphase winding, the one or more processors are configured to: in response to determining that a high frequency ripple current within the power system has reached a threshold, switch from implementing a first synthesis control scheme to a second synthesis control scheme at least until a target condition is met, wherein, under the first synthesis control scheme, the second switching elements are controlled by the one or more processors to generate the second PWM signals so as to be offset in phase from one another, and wherein, under the second synthesis control scheme, the second switching elements are controlled by the one or more processors to generate the second PWM signals so as to be in phase with one another.
(271) 27. The power system of any preceding clause, wherein the electric machine is an axial flux rotating electric machine.
(272) 28. The power system of any preceding clause, wherein the electric machine is a radial flux rotating electric machine.
(273) 29. A non-transitory computer readable medium comprising computer-executable instructions, which, when executed by one or more processors of a power system, cause the one or more processors to: control first switching elements of a power converter system to generate one or more first pulse width modulated (PWM) signals so as to render a first common mode signal, the

first PWM signals being generated based at least in part on one or more voltage commands associated with a first multiphase winding of an electric machine; and control second switching elements of the power converter system to generate one or more second PWM signals so as to render a second common mode signal, the second PWM signals being generated based at least in part on one or more voltage commands associated with a second multiphase winding of the electric machine, the second common mode signal having opposite polarity with respect to the first common mode signal, the first multiphase winding and the second multiphase winding being electrically opposite in phase with respect to one another.
(274) 30. The non-transitory computer readable medium of any preceding clause, wherein the second common mode signal has a same waveform as the first common mode signal.
(275) 31. The non-transitory computer readable medium of any preceding clause, wherein in generating the one or more second PWM signals, the one or more processors are configured to: shift a polarity of each one of the one or more second PWM signals.
(276) 32. The non-transitory computer readable medium of any preceding clause, wherein the first multiphase winding and the second multiphase winding are associated with a same channel of the power system.
(277) 33. The non-transitory computer readable medium of any preceding clause, wherein the first multiphase winding is associated with a first channel of the power system and the second multiphase winding is associated with a second channel of the power system.
(278) 34. The non-transitory computer readable medium of any preceding clause, wherein the electric machine is an axial flux rotating electric machine.
(279) 35. The non-transitory computer readable medium of any preceding clause, wherein the electric machine is a radial flux rotating electric machine.

Claims

1. A power system, comprising: a power converter system having first switching elements and second switching elements; an electric machine, comprising: a first multiphase winding electrically coupled with the first switching elements, the first multiphase winding comprising a first set of windings and a second set of windings connected in series; and a second multiphase winding electrically coupled with the second switching elements, the second multiphase winding comprising a first set of windings and a second set of windings connected in series, and wherein the first multiphase winding and the second multiphase winding are electrically opposite in phase with respect to one another; and one or more processors configured to: control the first switching elements to generate one or more first pulse width modulated (PWM) signals so as to render a first common mode signal, the first PWM signals being generated based at least in part on one or more voltage commands associated with the first multiphase winding; detect a failure condition associated with the second multiphase winding; and control, in response to the detected failure condition associated with the second multiphase winding, the second switching elements to synthesize a second common mode signal and generate one or more second PWM signals so as to render the second common mode signal, the second PWM signals being generated based at least in part on one or more voltage commands associated with the second multiphase winding, the second common mode signal having opposite polarity with respect to the first common mode signal.
2. The power system of claim 1, wherein the first set of windings and the second set of windings of the first multiphase winding are arranged in an interweaved contra-phase winding pair arrangement, and wherein the first set of windings and the second set of windings of the second multiphase winding are arranged in an interweaved contra-phase winding pair arrangement.
3. The power system of claim 1, wherein the electric machine defines a radial direction, wherein the first set of windings and the second set of windings of the first multiphase winding are positioned opposite one another along the radial direction, and wherein the first set of windings and

the second set of windings of the second multiphase winding are positioned opposite one another along the radial direction.

4. The power system of claim 1, wherein the electric machine defines a circumferential direction, wherein the first set of windings of the first multiphase winding is positioned between the first set of windings and the second set of windings of the second multiphase winding along the circumferential direction, and wherein the second set of windings of the first multiphase winding is positioned between the first set of windings and the second set of windings of the second multiphase winding along the circumferential direction.

5. The power system of claim 1, wherein the electric machine has a stator defining sectors, and wherein the first set of windings and the second set of windings of the first multiphase winding and the first set of windings and the second set of windings of the second multiphase winding are each wound within a respective one of the sectors of the stator.

6. The power system of claim 1, further comprising: a first gate driver in operative communication with the first switching elements, the first gate driver being configured to drive the first switching elements to generate the one or more first PWM signals so as to render the first common mode signal; and a second gate driver in operative communication with the second switching elements, the second gate driver being configured to drive the second switching elements to generate the one or more second PWM signals so as to render the second common mode signal.

7. The power system of claim 1, wherein the second common mode signal has a same waveform as the first common mode signal.

8. The power system of claim 1, wherein the first switching elements and the second switching elements are embodied in a single power converter.

9. The power system of claim 1, wherein the first switching elements and the second switching elements are connected in series.

10. The power system of claim 1, wherein in generating the one or more second PWM signals, the one or more processors are configured to: shift a polarity of each one of the one or more second PWM signals.

11. The power system of claim 1, wherein the first multiphase winding and the second multiphase winding are associated with a same channel of the power system.

12. The power system of claim 1, wherein the first multiphase winding is associated with a first channel of the power system and the second multiphase winding is associated with a second channel of the power system.

13. The power system of claim 1, wherein the electric machine has a stator defining a first sector, a second sector, a third sector, and a fourth sector, and wherein the first set of windings of the first multiphase winding is wound within the first sector, the second set of windings of the first multiphase winding is wound within the second sector, the first set of windings of the second multiphase winding is wound within the third sector, and the second set of windings of the second multiphase winding is wound within the fourth sector.

14. The power system of claim 1, wherein the electric machine has a rotor and a stator, and wherein the rotor is mechanically coupled with a rotating component of a gas turbine engine.

15. The power system of claim 1, wherein, in response to a detected failure condition associated with the power converter system, the one or more processors are configured to: control the second switching elements in one of a single-leg operation mode or a reduced leg operation mode to render the second common mode signal.

16. A method, comprising: receiving, by one or more processors, one or more voltage commands associated with a first multiphase winding of an electric machine, the first multiphase winding comprising a first set of windings and a second set of windings connected in series and being electrically coupled with first switching elements of a power converter system; receiving, by the one or more processors, one or more voltage commands associated with a second multiphase winding of the electric machine, the second multiphase winding comprising a first set of windings

and a second set of windings connected in series and being electrically coupled with second switching elements of the power converter system, the first multiphase winding and the second multiphase winding being electrically opposite in phase with respect to one another; controlling, by the one or more processors, the first switching elements to generate one or more first pulse width modulated (PWM) signals so as to render a first common mode signal, the first PWM signals being generated based at least in part on the one or more voltage commands associated with the first multiphase winding; detecting a failure condition associated with the second multiphase winding; and controlling, by the one or more processors, the second switching elements, in one of a single-leg operation mode or a reduced leg operation mode and in response to detecting the failure condition, to generate one or more second PWM signals so as to render a second common mode signal, the second PWM signals being generated based at least in part on the one or more voltage commands associated with the second multiphase winding, the second common mode signal having opposite polarity with respect to the first common mode signal.

17. The method of claim 16, further comprising: detecting a failure condition associated with the second multiphase winding; and controlling the second switching elements to synthesize the second common mode signal.

18. A power system, comprising: a power converter system having first switching elements, second switching elements, third switching elements, and fourth switching elements, wherein the first switching elements and the second switching elements are connected in series, and wherein the third switching elements and the fourth switching elements are connected in series; an electric machine, comprising: a first multiphase winding comprising a plurality of windings and being electrically coupled with the first switching elements; a second multiphase winding comprising a plurality of windings and being electrically coupled with the second switching elements, the first multiphase winding and the second multiphase winding being electrically opposite in phase with respect to one another; a third multiphase winding comprising a plurality of windings and being electrically coupled with the third switching elements; and a fourth multiphase winding comprising a plurality of windings and being electrically coupled with the fourth switching elements, the fourth multiphase winding being electrically opposite in phase with respect to the third multiphase winding; and one or more processors configured to: control the first switching elements to generate one or more first pulse width modulated (PWM) signals so as to render a first common mode signal, the first PWM signals being generated based at least in part on one or more voltage commands associated with the first multiphase winding; and control the second switching elements to generate one or more second PWM signals so as to render a second common mode signal, the second PWM signals being generated based at least in part on one or more voltage commands associated with the second multiphase winding, the second common mode signal having opposite polarity with respect to the first common mode signal.